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Section: 01

SOFTWARE PROJECT REPORT

Travel Management System

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Chapter 1: Introduction

1.1 Project Overview

User Story

Travel planning often requires users to switch between multiple applications for transportation booking, accommodation reservations, restaurant discovery, navigation, budgeting, weather updates, and emergency assistance. This fragmented experience makes trip planning time-consuming and inconvenient. Travelers also struggle to receive personalized recommendations, real-time safety support, and intelligent travel planning based on their preferences.

The **AI-Powered Travel Management System (AITMS)** is designed as an all-in-one platform that integrates every essential travel service into a single application. By leveraging Artificial Intelligence, the system generates personalized itineraries, recommends hotels and restaurants, predicts travel expenses and transportation fares, and adjusts travel plans according to weather conditions. Additional features such as emergency SOS services, travel partner matching, digital travel journals, and offline navigation enhance convenience, safety, and overall travel experience.

Traveler's Perspective

As a traveler, I want to plan and manage my entire journey from one platform so that I can easily book transportation, hotels, restaurants, manage my budget, receive AI-powered recommendations, stay safe during emergencies, and document my travel experiences without switching between multiple applications.

Travel Agent's Perspective

As a travel agent, I want to create tour packages, manage customer bookings, upload promotional offers, and provide customer support through a centralized platform so that I can efficiently serve travelers and expand my business.

Administrator's Perspective

As an administrator, I want to manage users, verify travel agents, monitor bookings, handle complaints, and analyze platform performance so that the system remains secure, reliable, and efficient for all stakeholders.

1.2 Objective and Scope

Objective

The purpose of the AI-Powered Travel Management System is to provide an intelligent, secure and user-friendly travel platform that simplifies every stage of a traveler's journey. The system integrates transportation booking, accommodation, restaurant discovery, navigation, budgeting, emergency support and AI-based travel planning into a single application, reducing the need for multiple independent services.

Scope

The system provides services for travelers, travel agents and administrators.

Its major functionalities include:

- User registration and authentication
- Transportation booking
- Hotel and accommodation booking
- Restaurant reservation

- AI-based travel itinerary generation
- Personalized travel recommendations
- Smart budget management
- Interactive maps and navigation
- Weather forecasting
- Emergency SOS services
- Community and travel partner matching
- Digital travel journal generation
- Administrative monitoring and analytics

The platform is designed for both web and mobile users and supports future integration with external travel services and APIs.

1.3 Stakeholders

The stakeholders of the AI-Powered Travel Management System are categorized into primary and secondary stakeholders.

Primary Stakeholders

- Travelers
- Travel Agents
- System Administrators

Secondary Stakeholders

- Hotels and Resorts
- Transportation Providers
- Restaurants

- Tour Guides
- Payment Gateway Providers
- Emergency Services (Police, Hospitals)
- Government Tourism Authorities
- Third-party API Providers

1.4 Technology Stack

Category	Technology
Frontend	React.js , Vite, Tailwind CSS
Backend	PHP
Database	MySQL XAMPP
AI	OpenAI API / Machine Learning Models
Maps	Google Maps API
Authentication	JWT Authentication
Payment	SSLCommerz / Stripe (Future Integration)

Version Control	Git & GitHub
Deployment	Vercel, Render
UI Design	Figma
Diagram Design	Draw.io

Chapter 2: Software Requirements Specification & Analysis

2.1 Stakeholder Needs & Analysis

The success of the AI-Powered Travel Management System depends on understanding the needs of all stakeholders involved. The system is designed to provide a unified platform that addresses the requirements of travelers, travel agents, administrators, and supporting organizations. Stakeholder analysis helps identify expectations and ensures that the system delivers value to all users.

Primary Stakeholders

Stakeholder	Needs
Traveler	Easy registration, transportation and accommodation booking, AI-based itinerary planning, budget management, restaurant recommendations, emergency assistance, secure document storage and personalized travel recommendations.
Travel Agent	Create and manage tour packages, manage bookings, upload promotional offers, communicate with customers and monitor package performance.

Administrator	Manage users and travel agents, verify listings, handle complaints, monitor system performance and analyze platform activities through dashboards.
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Table 2.1: Primary Stakeholders and Their Needs

Secondary Stakeholders

Stakeholder	Needs
Hotels & Resorts	Receive accommodation bookings and manage room availability.
Transportation Providers	Provide schedules, seat availability, and fare information.
Restaurants	Accept reservations and food pre-orders from travelers.
Tour Guides	Offer guiding services and interact with travelers.
Emergency Services	Receive SOS alerts and assist travelers during emergencies.
Payment Gateway Providers	Process secure online payments.
Government Tourism Authorities	Promote tourism and provide verified travel information.
Third-party API Providers	Supply weather, maps, navigation, and travel-related services.

Table 2.2: Secondary Stakeholders and Their Needs

Stakeholder Analysis

The collected stakeholder requirements indicate that travelers expect a single platform capable of planning, booking, managing, and enhancing their travel experience. Travel agents require efficient tools to manage services and customers, while administrators need monitoring and management capabilities to ensure platform reliability and security. These requirements guided the development of both the functional and non-functional requirements of the proposed system.

2.2 Requirement Elicitation Methods

To gather the project requirements, multiple requirement elicitation techniques were employed. These methods helped identify user expectations, existing system limitations, and opportunities for improvement.

Interviews

Interviews were conducted with participants from different age groups, including travelers, travel agents, and tour guides. The discussions focused on travel planning experiences, booking preferences, safety concerns, and expectations from an AI-powered travel platform.

Interview Topics

- Common challenges during travel
- Preferred booking methods
- Desired AI-based features
- Safety and emergency requirements
- Budget management expectations

Interview Recordings:

Drive links: https://drive.google.com/file/d/1-SzvfJxPrVu0b2ycSGm5_DMhLN0Ei_HW/view?usp=drive_link

<https://drive.google.com/file/d/1Cutfb5jH8pGgAFUjjTum00Wx8H4Fu1t7/view?usp=sharing>

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<https://drive.google.com/file/d/13zXZmZnK1v2cIn8hV1mLHtY35cJaN46Q/view?usp=sharing>

https://drive.google.com/file/d/1WMVVpEp_W-2sCvleRkhQXSNk7To-MbP6/view?usp=sharing

Surveys

An online survey was conducted to understand user preferences and travel behavior.

Sample Survey Questions

- What are your most common travel difficulties?
- Which booking method do you prefer?
- How important are AI-based travel recommendations?
- What safety concerns do you face while traveling?
- Which travel services do you use most frequently?

The survey responses helped identify common pain points and prioritize system features.

Observation

Several existing travel-related applications and services were analyzed to understand their strengths and limitations.

Booking Platforms

- Require users to switch between multiple applications.
- Limited AI-based personalization.
- Limited integrated emergency support.

Navigation Applications

- Heavy dependence on internet connectivity.

- Offline navigation support is limited.
- Difficult to use in remote areas.

Travel Blogs

- Information may be outdated or inaccurate.
- Lack of verification and personalization.

Food Delivery Applications

- Focus mainly on ordering food.
- Do not recommend local restaurants based on travel plans or nearby attractions.

Summary of Requirement Elicitation

The collected information revealed several important user needs:

- A single platform for complete travel management.
- AI-assisted travel planning and recommendations.
- Enhanced traveler safety through emergency services.
- Smart budget planning and expense tracking.
- Integrated transportation, accommodation, restaurant, and community services.

These findings formed the foundation for the system requirements presented in the following sections.

2.3 List of Requirements

2.3.1 Functional Requirements (FR)

Functional requirements describe the services and operations that the AI-Powered Travel Management System must provide to its users.

FR-01 User Management

The system shall:

- Allow users to register and log in securely.
- Allow users to manage and update profile information.
- Maintain travel history records.
- Allow users to manage saved destinations, travel memories, and wishlists.
- Provide real-time chat functionality.
- Allow users to specify travel preferences, budget, medical conditions, food allergies, preferred destinations, and additional service requirements.

FR-02 Transportation Management

The system shall:

- Allow booking of buses, trains, flights, ferries, ride-sharing services, and car rentals.
- Support seat selection.
- Support ticket cancellation and rescheduling.
- Display real-time fare and availability information.
- Predict future fare changes using AI.

FR-03 Accommodation Management

The system shall:

- Support hotel, resort, hostel, and homestay bookings.
- Display room availability and pricing.
- Support reviews and ratings.
- Recommend accommodations based on user preferences.

FR-04 Restaurant & Essentials Management

The system shall:

- Support restaurant reservations.
- Support food pre-ordering.
- Recommend nearby restaurants.
- Support grocery and daily essentials pre-ordering.
- Filter services by price, distance, and dietary preferences.

FR-05 AI Travel Planner & Assistant

The system shall:

- Generate AI-based travel itineraries.
- Generate travel cost estimates.
- Provide personalized recommendations based on user preferences, budget, age, mood, and travel history.
- Export itineraries as PDF and save them to user profiles.
- Support voice and text interaction.
- Provide AI-powered travel assistance and language translation.
- Match travelers with similar interests and destinations.
- Detect potential travel risks such as unsafe areas, severe weather, scams, and disease outbreaks.

FR-06 Emergency & Safety Services

The system shall:

- Provide an SOS emergency button.

- Share live location during emergencies.
- Notify emergency contacts automatically.
- Display nearby hospitals, police stations, and emergency services.

FR-07 Business & Enterprise Services

The system shall:

- Support corporate travel booking and approval workflows.
- Manage event and conference travel.
- Support customized honeymoon packages.
- Support family travel planning with shared itineraries and expense management.

FR-08 Community Features

The system shall:

- Allow users to post travel-related content.
- Provide discussion forums.
- Support reviews and ratings.
- Allow users to share travel experiences, photos, and videos.

FR-09 Group Travel Management

The system shall:

- Allow users to create travel groups.
- Support shared itineraries.
- Support expense splitting.
- Support voting-based decision-making.

FR-10 Travel Document Management

The system shall:

- Store passports, visas, tickets, and hotel confirmations securely.
- Provide QR-code-based document access.
- Send document expiry reminders.

FR-11 Smart Budget Management

The system shall:

- Track transportation, accommodation, food, and shopping expenses.
- Provide budget alerts and overspending warnings.
- Generate spending reports.

FR-12 Weather & Navigation Services

The system shall:

- Display real-time weather information.
- Suggest itinerary changes based on weather forecasts.
- Support live navigation and offline maps.
- Optimize travel routes.

2.3.2 Non-Functional Requirements (NFR)

Non-functional requirements describe the quality attributes and operational constraints of the system.

NFR-01 Security

The system shall:

- Encrypt user passwords.

- Securely store personal documents.
- Implement role-based access control.
- Ensure secure authentication and authorization.

NFR-02 Performance

The system shall:

- Respond within 3 seconds under normal conditions.
- Support at least 10,000 concurrent users.
- Provide real-time updates for bookings and fares.

NFR-03 Reliability

The system shall:

- Maintain at least 99% availability.
- Perform daily automated backups.
- Recover critical data after failures.

NFR-04 Usability

The system shall:

- Provide an intuitive interface.
- Support mobile, tablet, and desktop devices.
- Support English and Bangla languages.

NFR-05 Scalability

The system shall:

- Support future integration with external APIs.

- Handle increasing users and transactions.
- Allow addition of new modules.

NFR-06 Maintainability

The system shall:

- Follow modular architecture.
- Allow easy updates and bug fixes.
- Maintain complete documentation.

NFR-07 Compatibility

The system shall:

- Support major web browsers.
- Support Android and iOS devices.
- Adapt to different screen sizes.

2.3.3 Extraordinary Requirements (Wow Factors)

To differentiate the proposed system from existing travel applications, several innovative features are incorporated.

- AI Smart Travel Planner
- AI Personalized Recommendation Engine
- AI Budget Fare Prediction

These advanced features provide a more personalized, intelligent, and secure travel experience compared to traditional travel management platforms.

2.4 Quality Function Deployment (QFD)

Quality Function Deployment (QFD) is a systematic technique used to translate customer needs into technical requirements. It ensures that the developed system satisfies stakeholder expectations by establishing relationships between customer requirements and engineering solutions.

2.4.1 Customer Requirements (CR)

Customer Requirements represent the expectations and needs of travelers, travel agents, and administrators.

ID	Customer Requirement
CR-01	A single platform for complete travel management.
CR-02	Easy transportation and accommodation booking.
CR-03	Personalized AI-based travel recommendations.
CR-04	Real-time weather and travel information.
CR-05	Smart budget planning and expense tracking.
CR-06	Easy discovery of restaurants and tourist attractions.
CR-07	Emergency assistance and traveler safety.
CR-08	Travel partner matching with similar interests.
CR-09	Efficient group travel planning.
CR-10	Secure storage of travel documents.
CR-11	Efficient management tools for travel agents.

CR-12	Administrative monitoring and analytics dashboard.
CR-13	Fast and responsive system performance.
CR-14	Mobile-friendly and bilingual interface.
CR-15	Intelligent and personalized travel experience through AI.

Table 2.3: Customer Requirements

2.4.2 Engineering Requirements (TR)

Engineering Requirements define the technical solutions required to satisfy the customer requirements.

ID	Engineering Requirement
TR-01	Secure authentication and authorization.
TR-02	Password encryption and secure document storage.
TR-03	Role-based access control.
TR-04	Transportation and accommodation booking modules.
TR-05	Hotel availability and reservation management.
TR-06	AI itinerary generation and recommendation engine.
TR-07	Real-time integration with weather, maps, and booking APIs.

TR-08	Interactive online and offline navigation.
TR-09	SOS emergency service with live location sharing.
TR-10	PDF itinerary and report generation.
TR-11	Responsive bilingual user interface.
TR-12	System response time below 3 seconds.
TR-13	Support for at least 10,000 concurrent users.
TR-14	99% system availability with daily backup.
TR-15	Integration with external APIs and future services.
TR-16	AI budget prediction and fare forecasting.
TR-17	Travel partner matching algorithm.
TR-18	Carbon footprint analysis module.

Table 2.4: Engineering Requirements

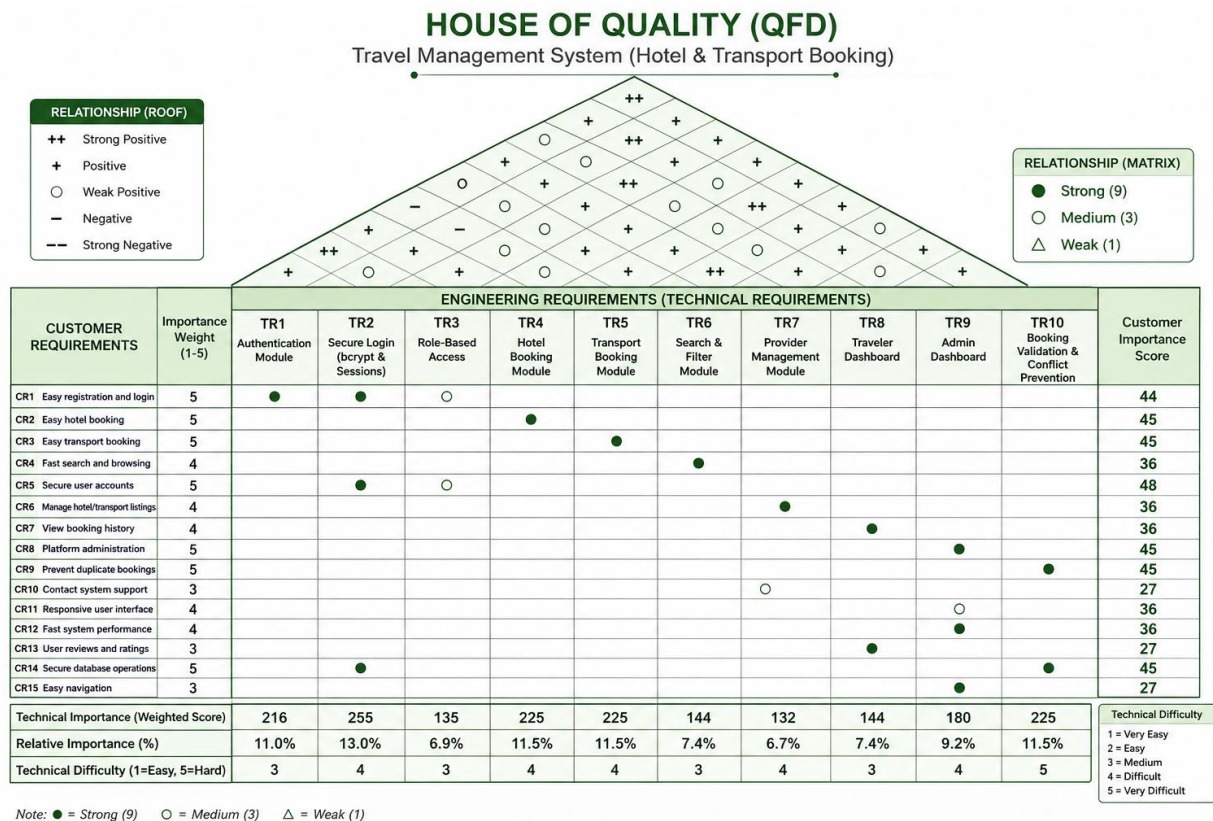
2.4.3 Quality Function Deployment Matrix (House of Quality)

The following matrix illustrates the relationship between customer requirements and engineering requirements. A **Strong (9)** relationship indicates direct support, **Medium (3)** indicates partial support, and **Weak (1)** indicates indirect support.

Customer Requirements	Engineering Requirements	Relationship
-----------------------	--------------------------	--------------

CR-01 Single travel platform	TR-04 Booking Modules	Strong (9)
CR-02 Easy booking	TR-04 Booking Modules	Strong (9)
CR-03 AI recommendations	TR-06 AI Recommendation Engine	Strong (9)
CR-04 Real-time information	TR-07 API Integration	Strong (9)
CR-05 Budget management	TR-16 AI Budget Prediction	Strong (9)
CR-06 Restaurant discovery	TR-07 API Integration	Medium (3)
CR-07 Traveler safety	TR-09 SOS & Live Location	Strong (9)
CR-08 Travel partner matching	TR-17 Matching Algorithm	Strong (9)
CR-09 Group travel	TR-17 Matching Module	Medium (3)
CR-10 Secure document storage	TR-02 Secure Storage	Strong (9)
CR-11 Travel agent management	TR-03 Role-Based Access	Strong (9)
CR-12 Admin analytics	TR-03 Admin Dashboard	Strong (9)
CR-13 Fast performance	TR-12 Performance Optimization	Strong (9)
CR-14 Mobile & bilingual support	TR-11 Responsive UI	Strong (9)
CR-15 Intelligent travel experience	TR-06 AI Engine	Strong (9)

Table 2.8: Quality Function Deployment (QFD) Matrix



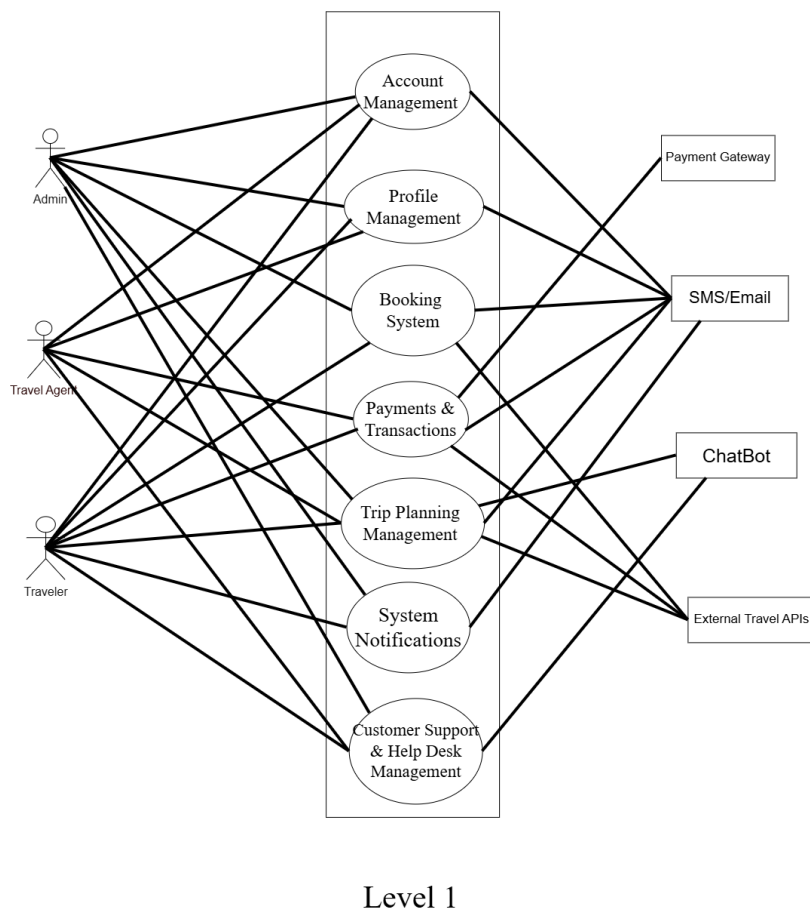
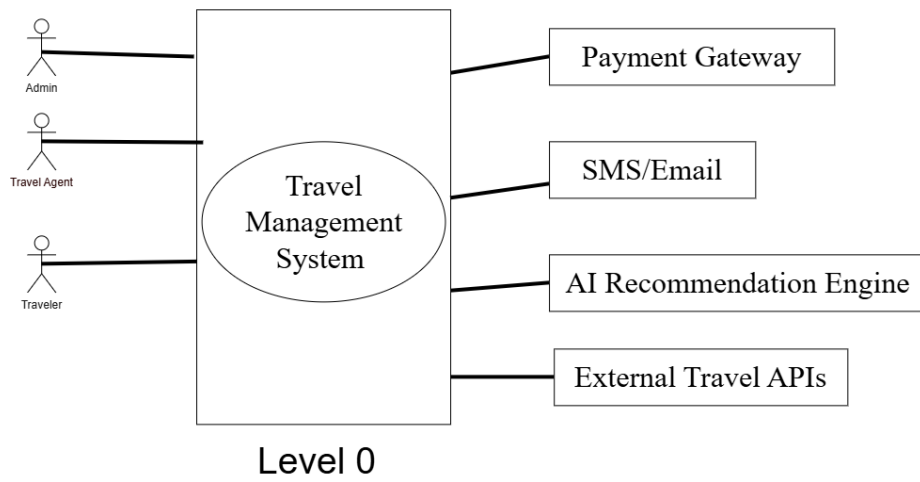
2.5 Requirements Modeling

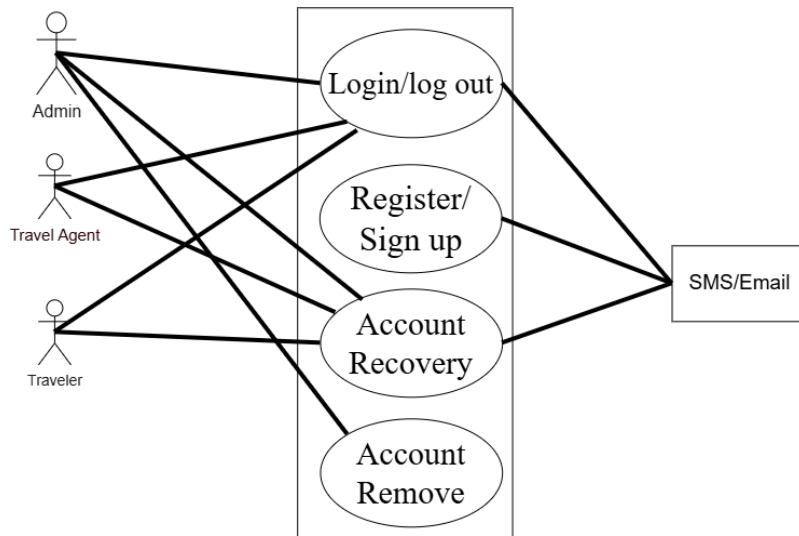
Requirements modeling provides a visual representation of the system's behavior and interactions. The following UML diagrams were developed to illustrate the functional workflow of the proposed system.

2.5.1 Use Case Diagram

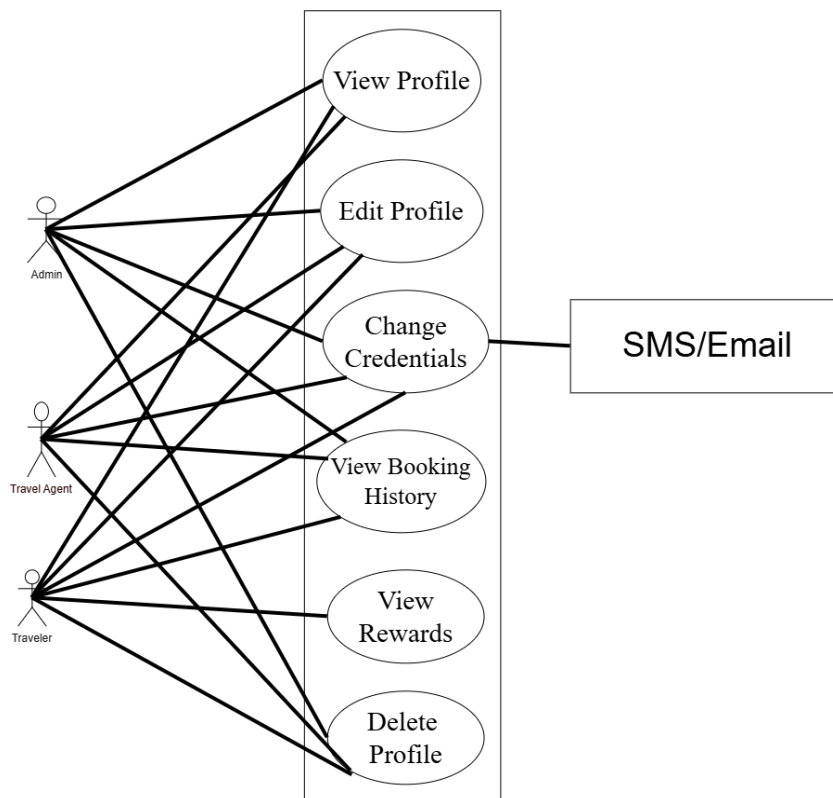
The Use Case Diagram illustrates the interactions between the primary actors (Traveler, Travel Agent, and Administrator) and the AI-Powered Travel Management System. It represents the major services provided by the system, including booking management, AI travel planning, emergency services, community features, and administrative functions.

Link: https://drive.google.com/file/d/1roI262yHi-1zN_EzLdU6YbFsTmYRVQRN/view?usp=sharing

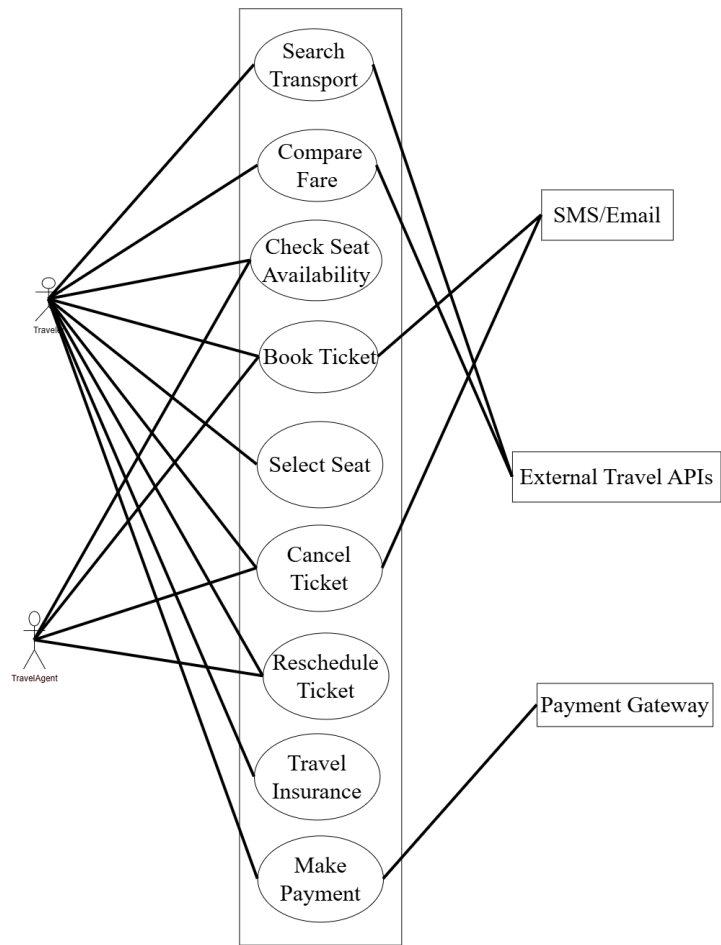




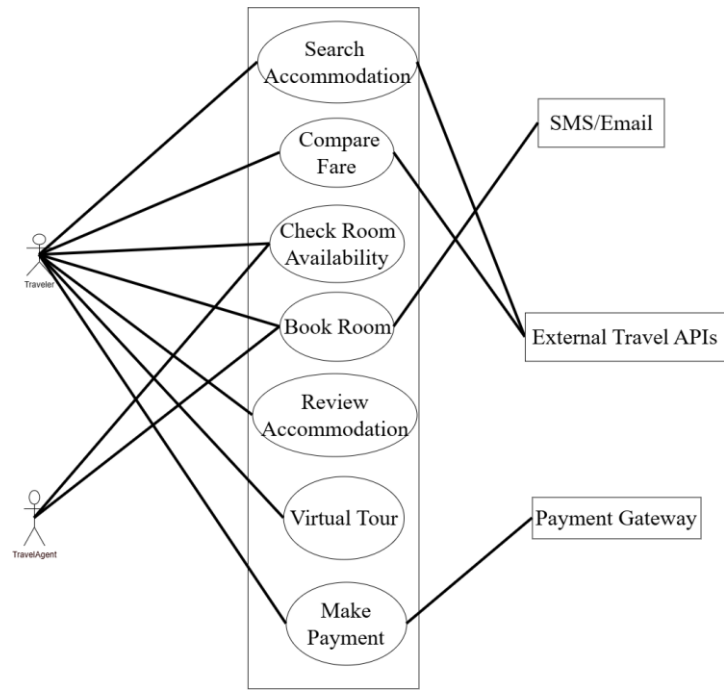
Level 1.1: Account Management



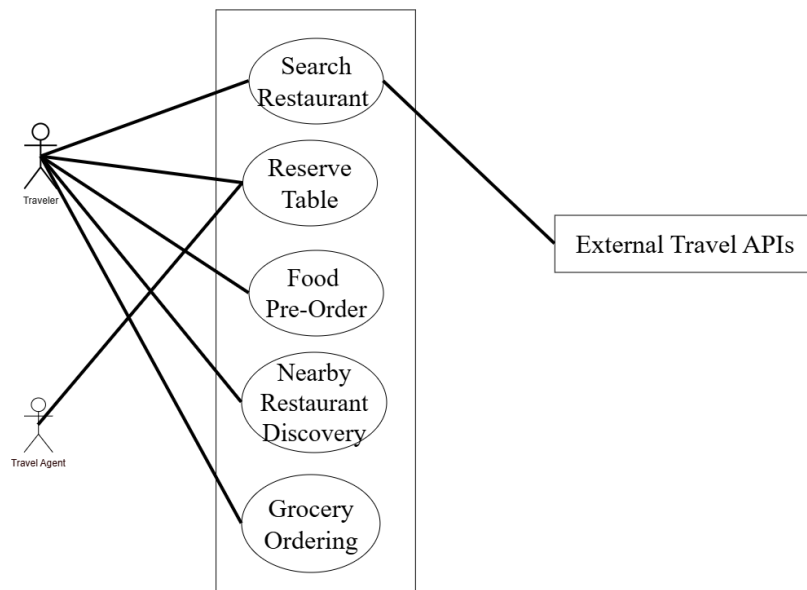
Level 1.2: Profile Management



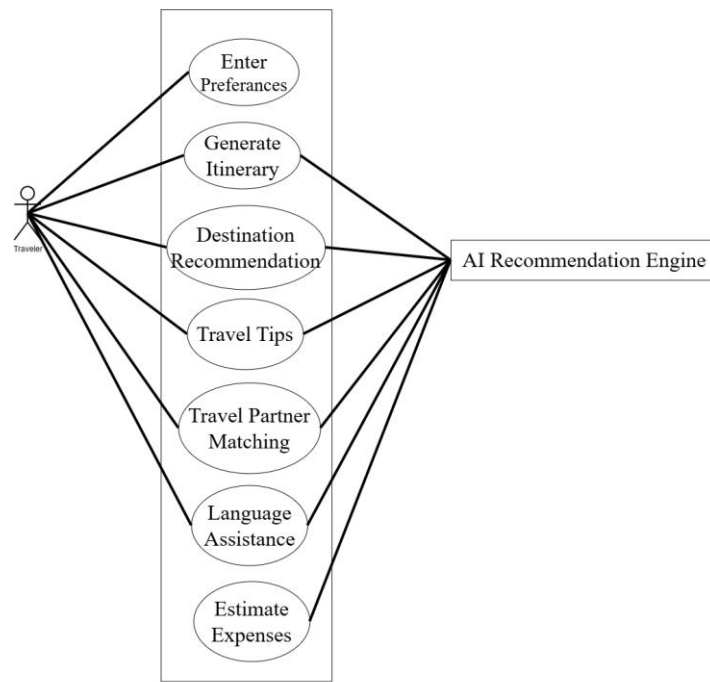
Transportation Management



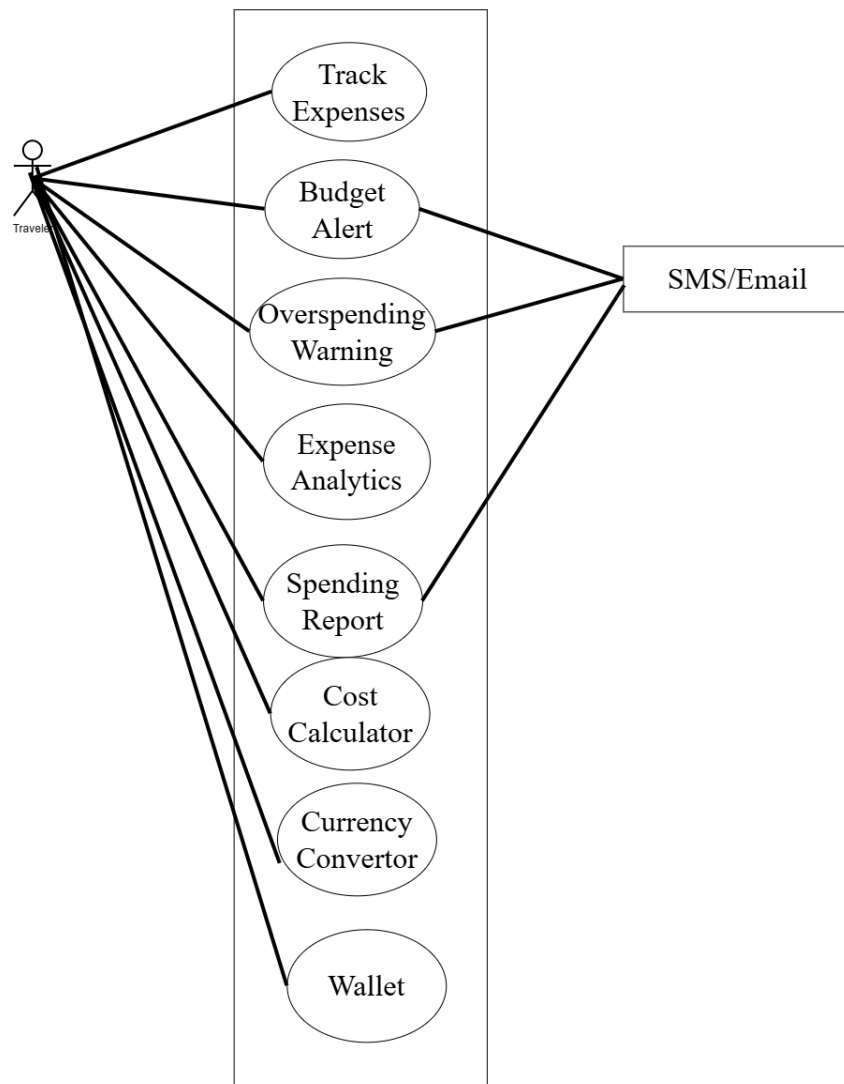
Accommodation Management



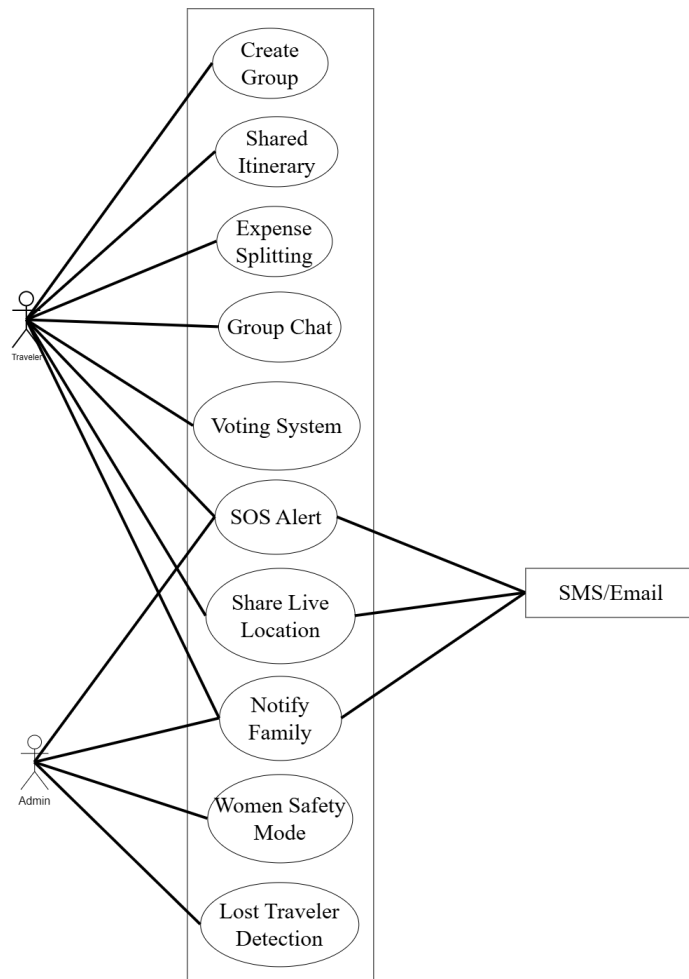
Restaurant & Essentials

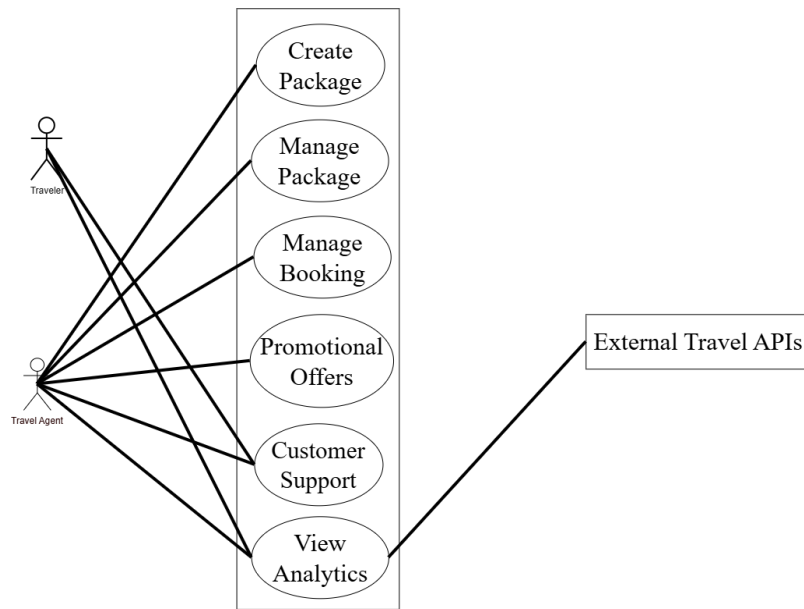


AI Travel Planning & Companion System

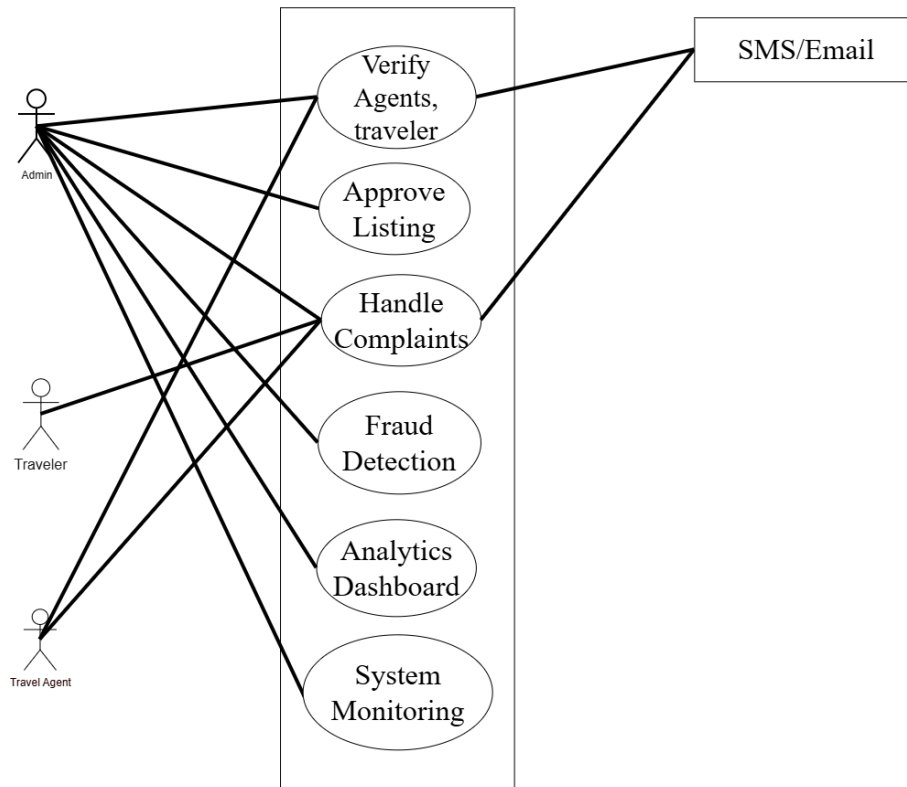


Budget Management

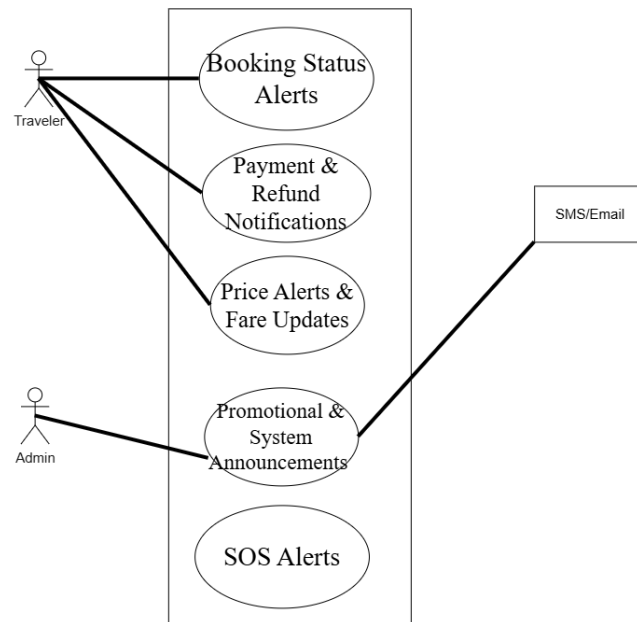




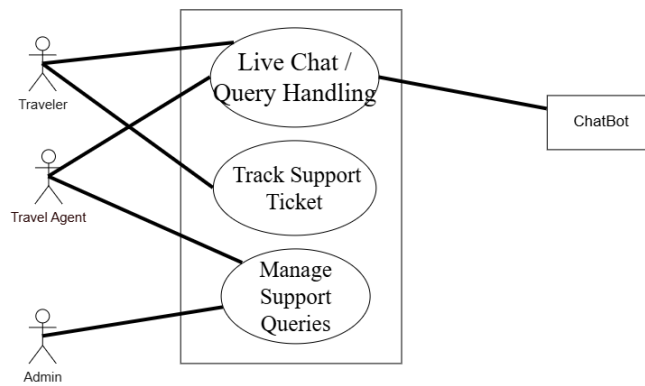
Travel Agent Management



Administrator Management



Level : System Notification

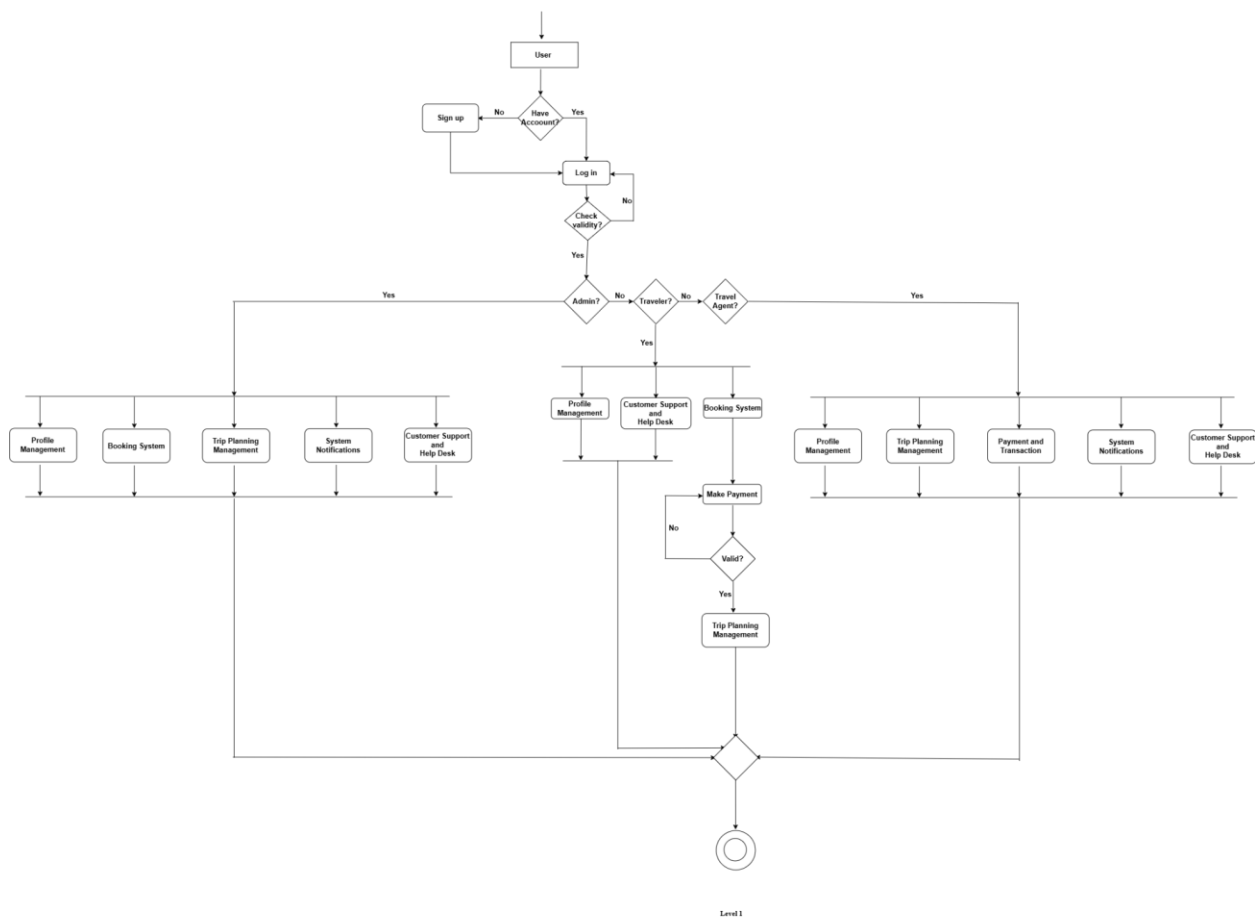


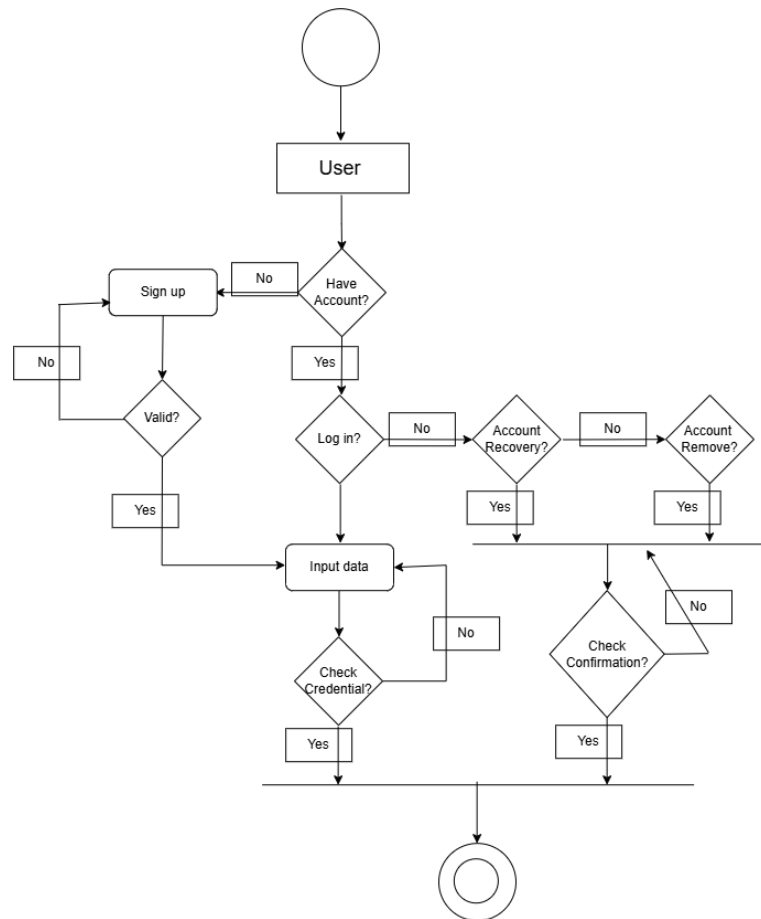
Level : Customer Support & Help Desk Management

2.5.2 Activity Diagram

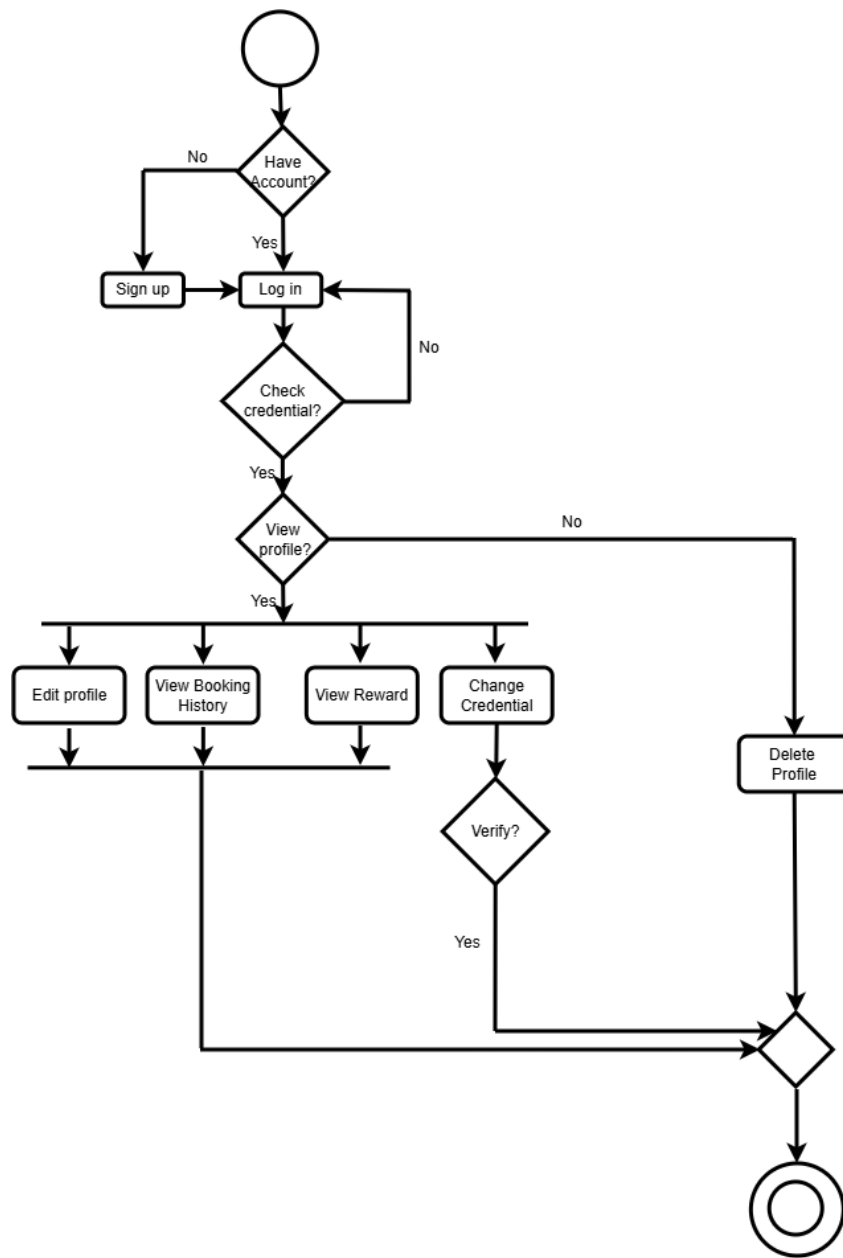
The Activity Diagram describes the workflow of the AI Smart Travel Planner. It begins with user authentication, followed by entering travel preferences such as destination, budget, duration, and interests. The AI engine analyzes the information, generates personalized recommendations, and produces an optimized travel itinerary that can be saved or exported.

Link: https://drive.google.com/file/d/1roI262yHi-1zN_EzLdU6YbFsTmYRVQRN/view?usp=sharing

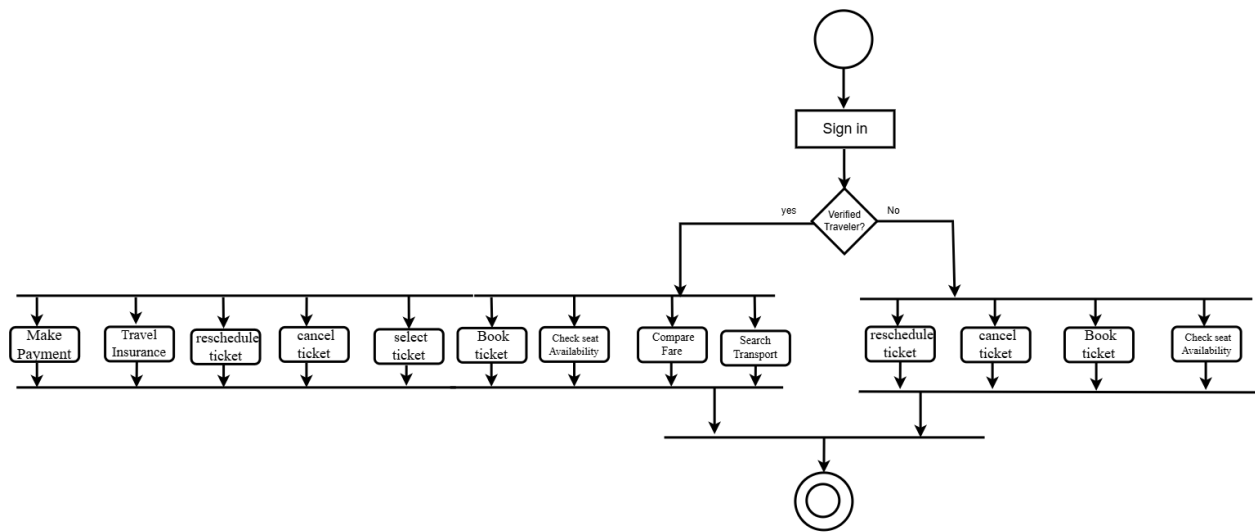




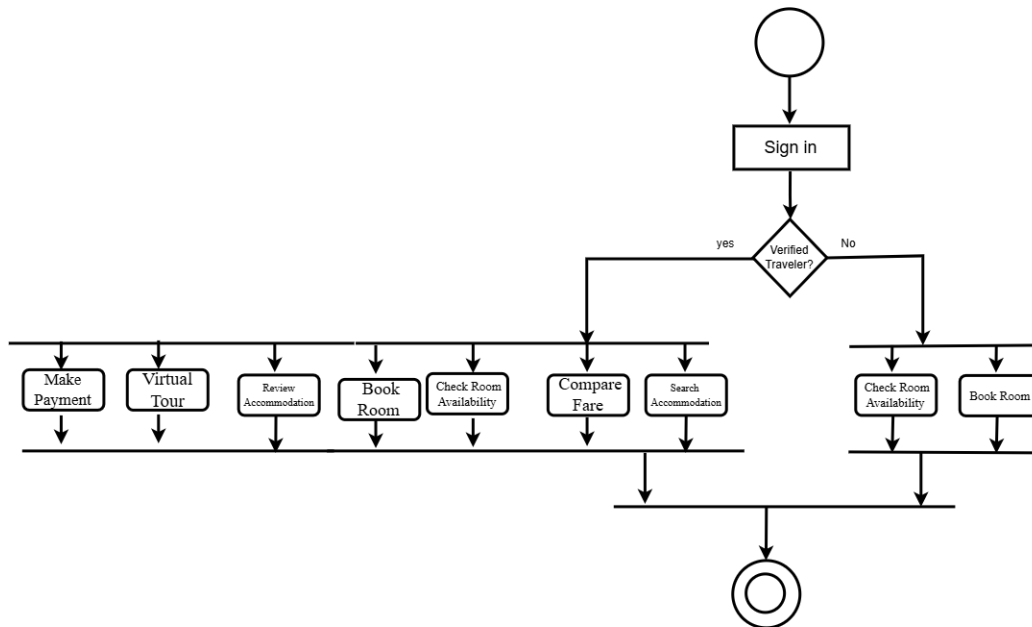
Level 1.1: Account Management



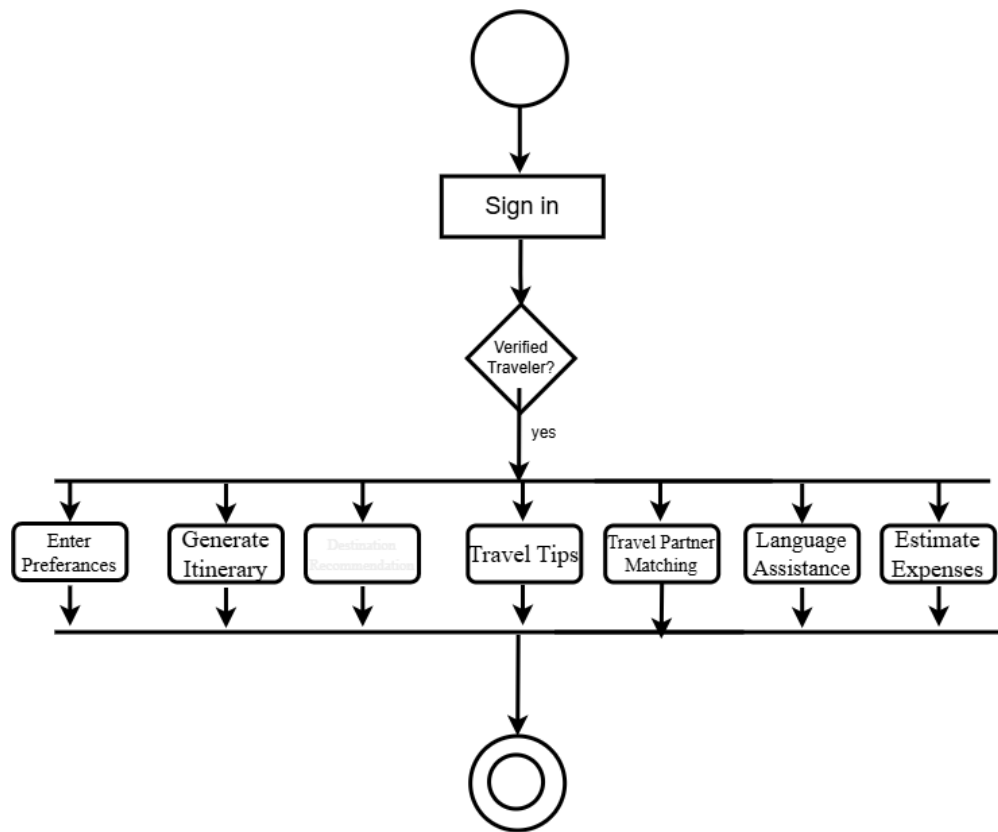
Level 1.2: Profile Management



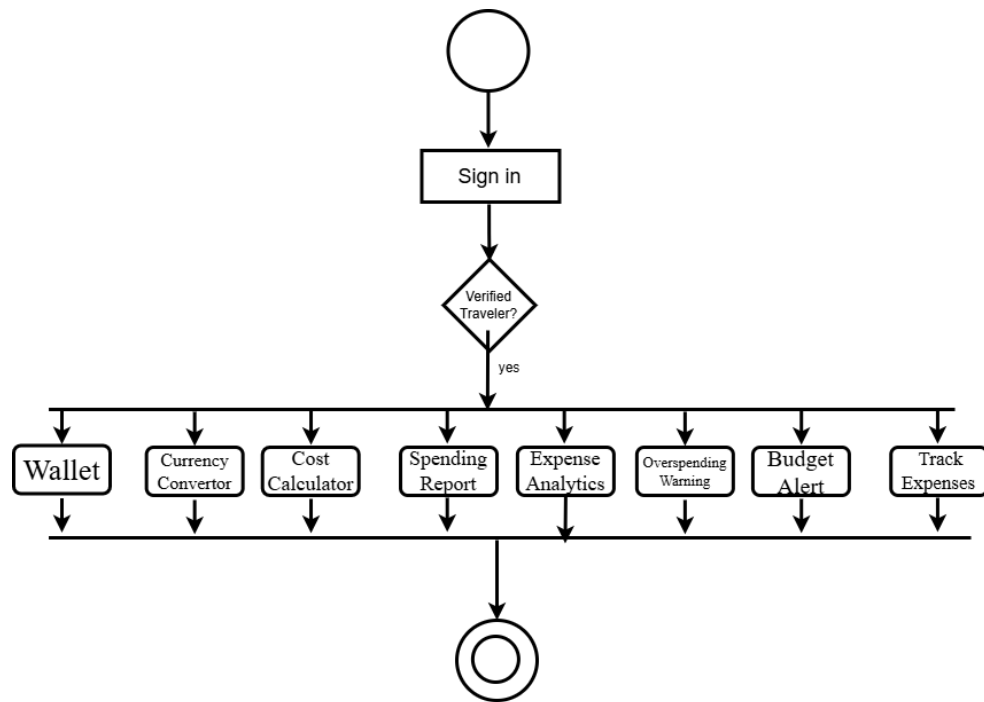
Level 1.3: Transportation Management



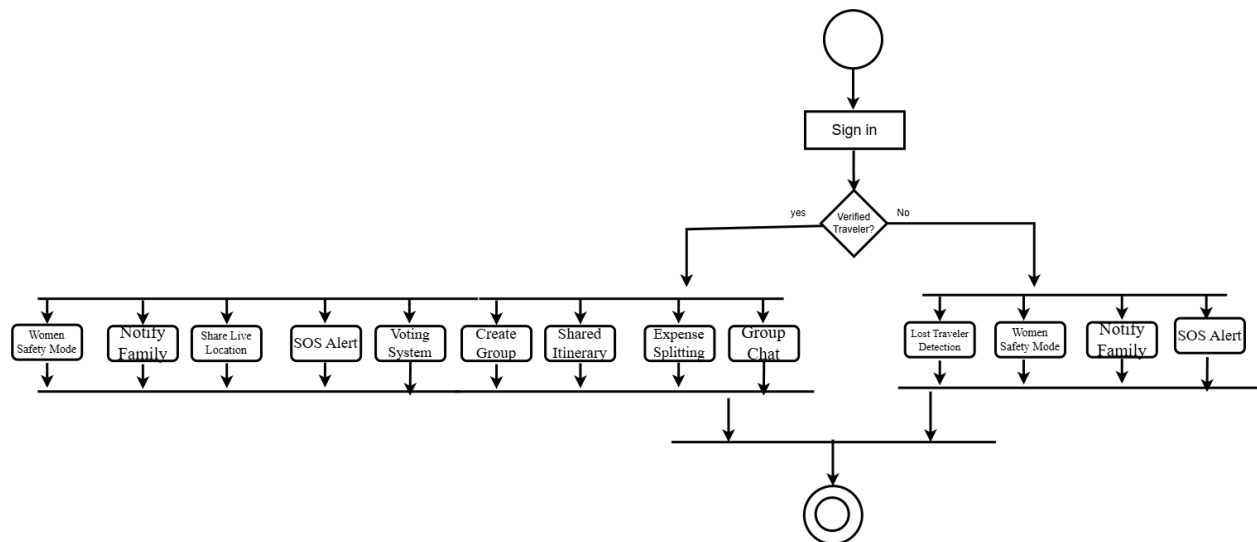
Level 1.4: Accommodation Management



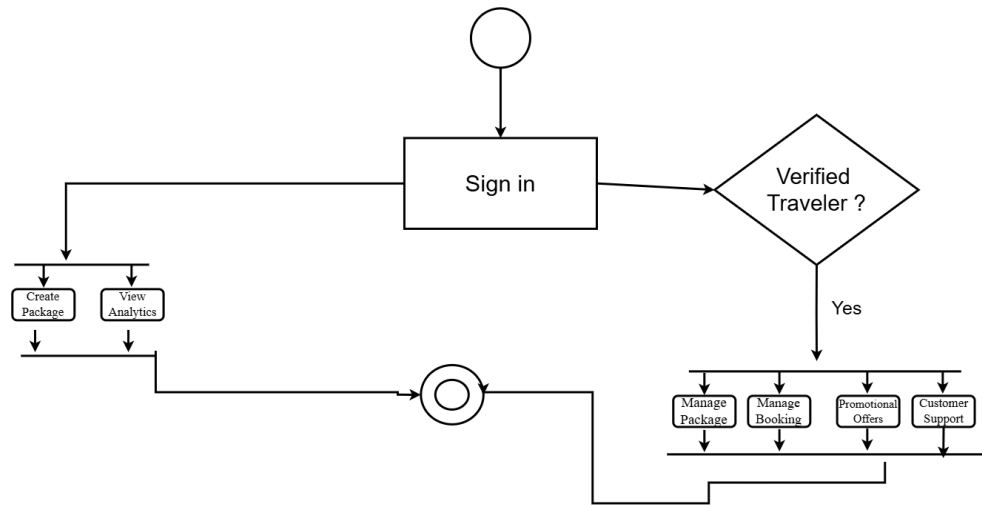
Level 1.6: AI Travel Planning & Companion System



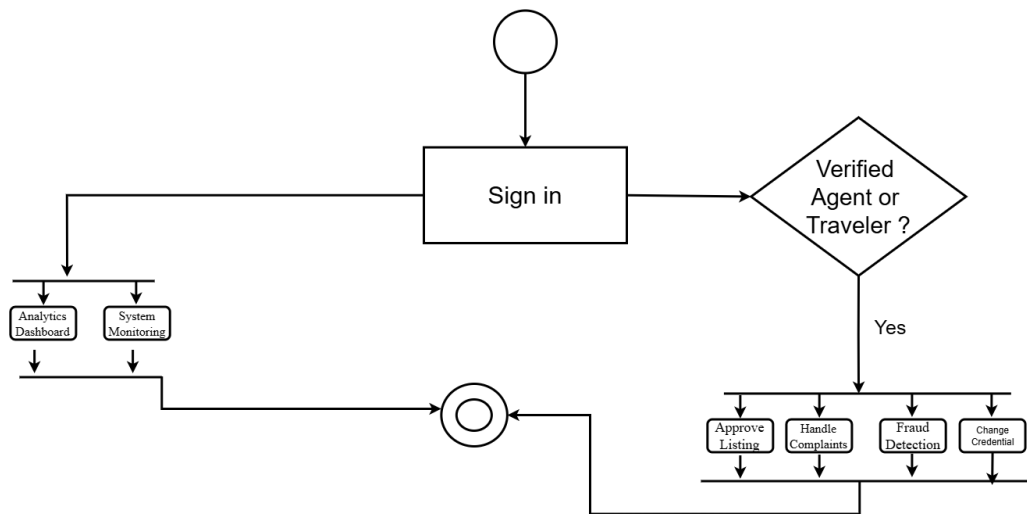
Level 1.7: Budget Management



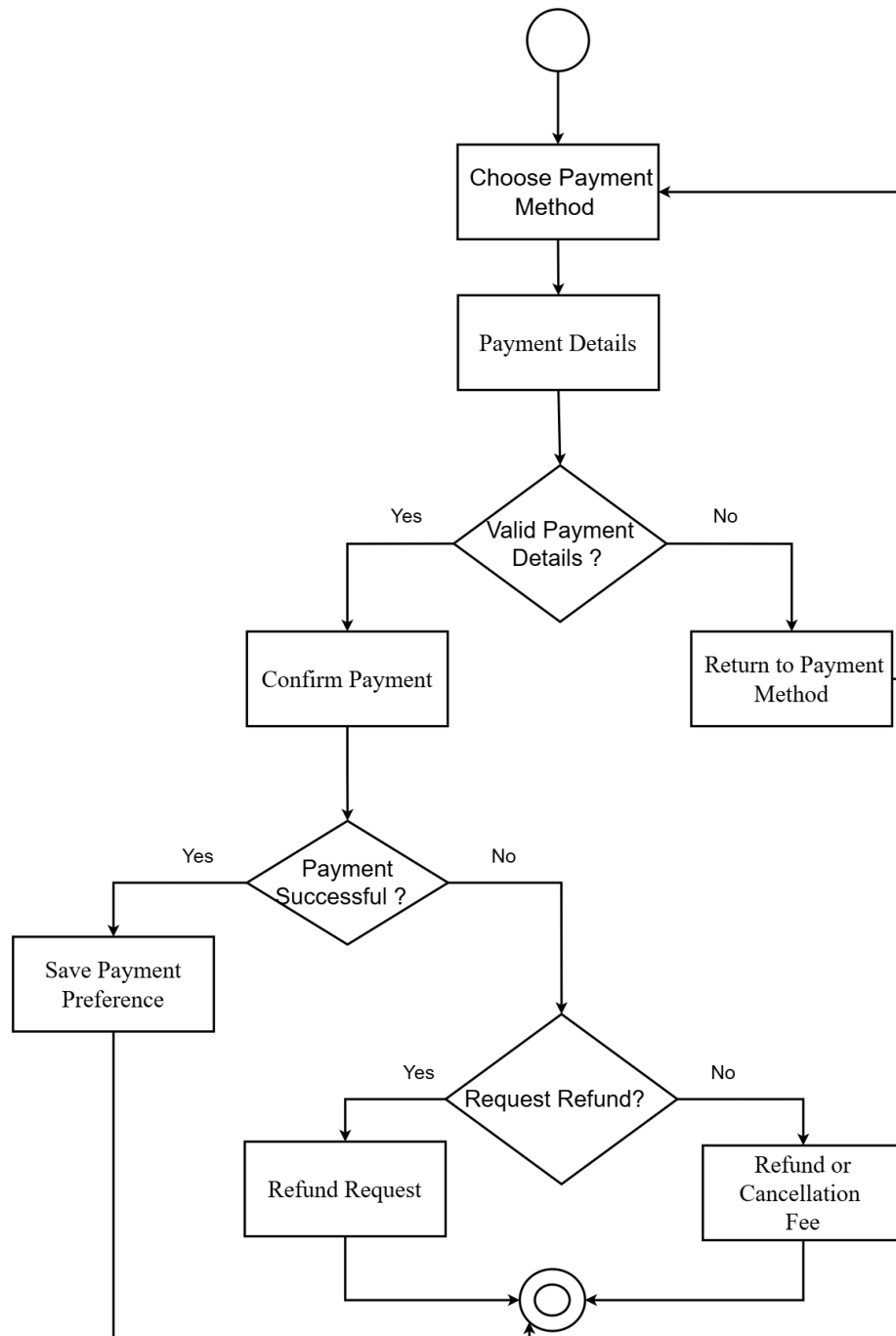
Level 1.8: Group Travel & Safety Management



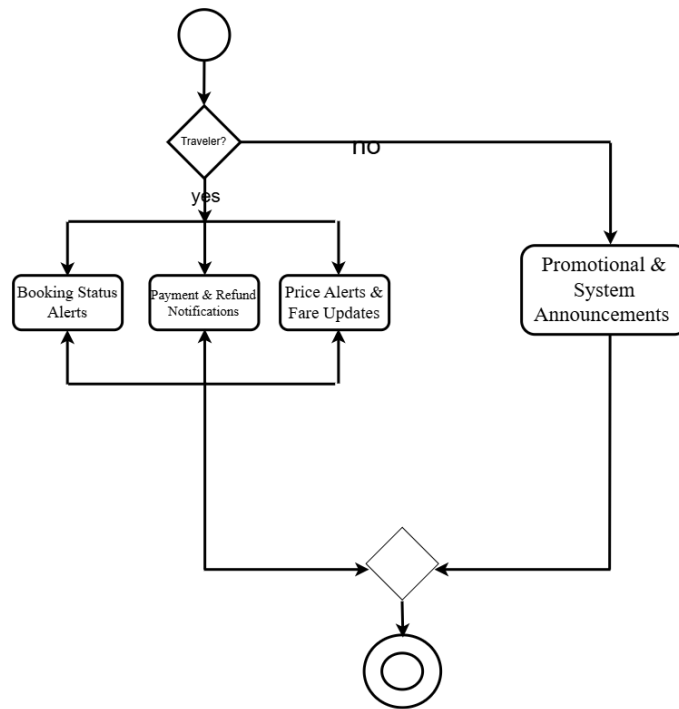
Level 1.9: Travel Agent Management



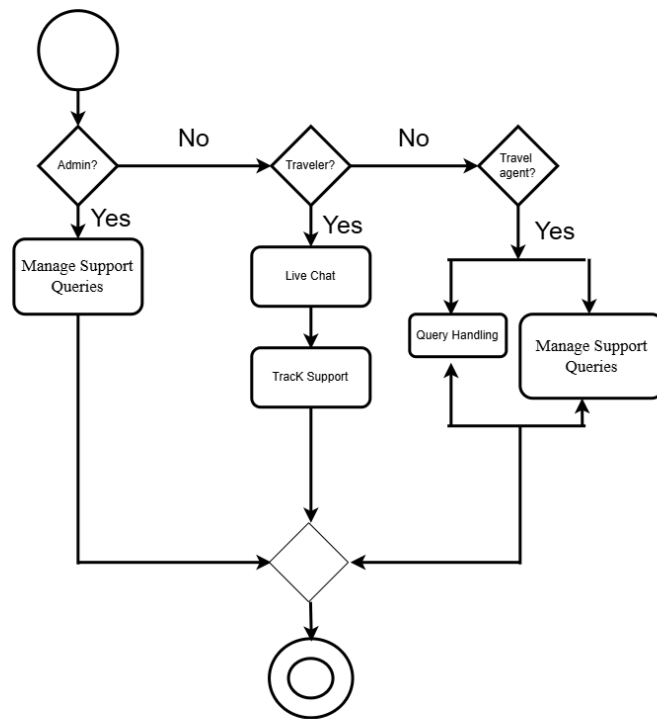
Level 2: Administrator
Management



Level 2.1: Payments & Transactions



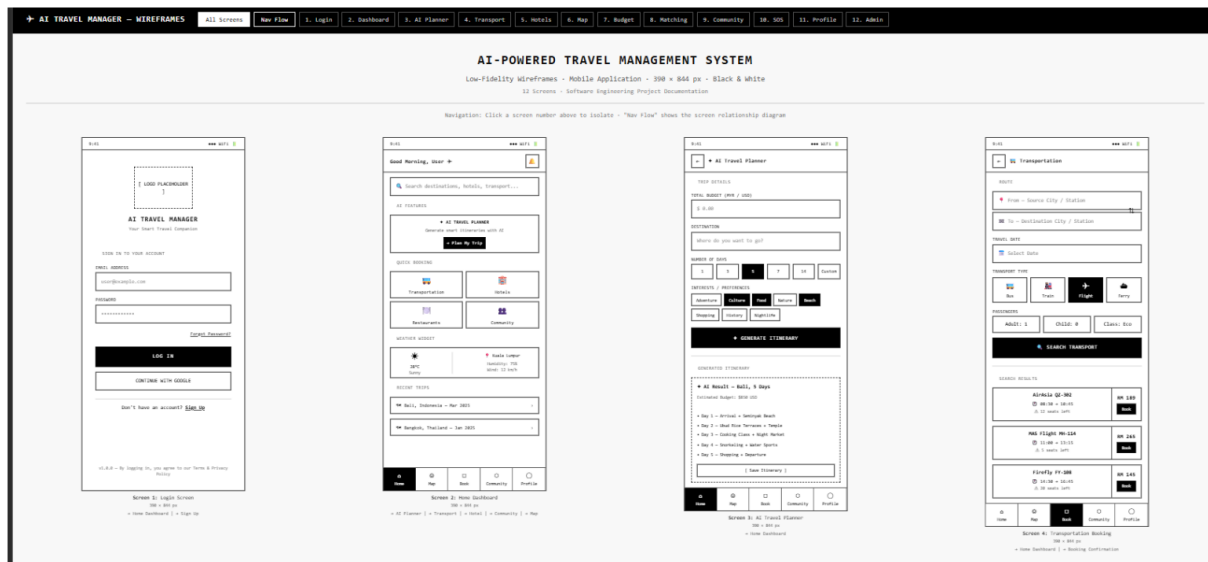
Level 2.2: System Notification

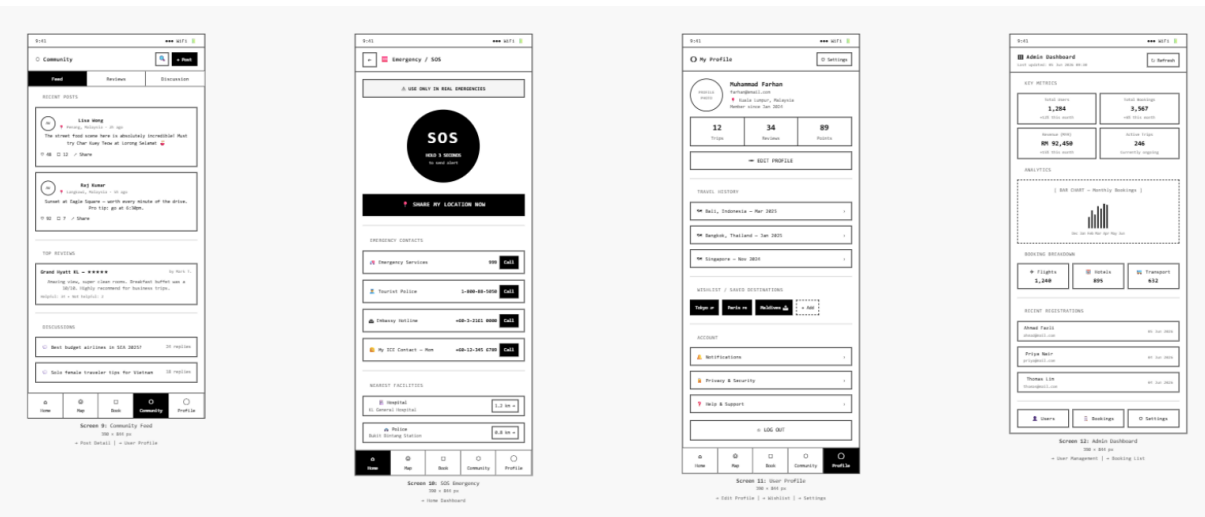


Level 2.3: Customer Support & Help Desk Management

2.5.3 User Interface Sketches

The UI sketches provide a preliminary visualization of the application's layout and navigation. They illustrate the arrangement of interface elements before detailed UI design is performed. The sketches include the Login Screen, Home Dashboard, AI Travel Planner, Transportation Booking, Hotel Booking, Emergency SOS, User Profile, and Admin Dashboard.





Chapter 3: Software Design

Software design transforms the system requirements into a structured architecture that guides implementation. This chapter presents the architectural design, component-level design, database design, and user interface design of the proposed AI-Powered Travel Management System.

3.1 Architectural Design

The architectural design illustrates the overall organization of the system and the interaction between users, software components, databases, and external services.

3.1.1 Architectural Context Diagram

The AI-Powered Travel Management System interacts with three primary actors: **Travelers**, **Travel Agents**, and **Administrators**. It also communicates with several external services, including payment gateways, Google Maps, weather services, emergency services, and AI recommendation engines.

The system receives user requests, processes them through the backend server and AI modules, stores and retrieves information from the database, and communicates with external APIs to provide real-time travel services.

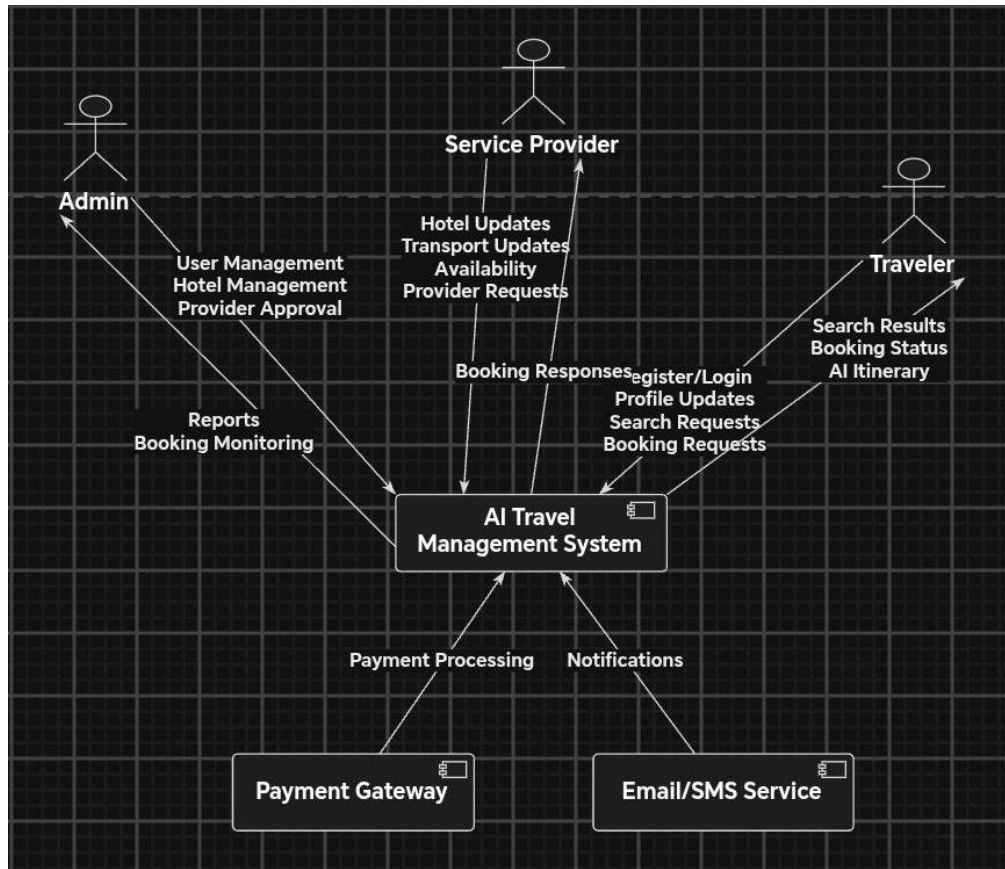


Figure 3.1: Architectural Context Diagram

3.1.2 Top-Level Component Diagram

The system is divided into several major components, each responsible for a specific functionality.

Main Components

1. User Management Module

Responsible for:

- Registration
- Login
- Authentication

- Profile Management
- Travel History
- Wishlist

2. Booking Management Module

Responsible for

- Transportation Booking
- Hotel Booking
- Restaurant Reservation
- Travel Documents

3. AI Module

Responsible for

- AI Travel Planner
- Recommendation Engine
- Budget Prediction
- Fare Prediction
- Weather-based Planning
- Travel Partner Matching

4. Emergency & Safety Module

Responsible for

- SOS

- Live Location Sharing
- Scam Alerts
- Women Safety Mode
- Lost Traveler Detection

5. Community Module

Responsible for

- Travel Feed
- Reviews
- Forums
- Group Travel
- Travel Journal

6. Administration Module

Responsible for

- User Management
- Agent Management
- Complaint Handling
- Analytics Dashboard
- Listing Verification

7. External Service Layer

Responsible for communication with

- Google Maps API
- Weather API
- Payment Gateway
- AI APIs
- Email & Notification Services

8. Database

Stores

- Users
- Bookings
- Hotels
- Restaurants
- Transportation
- Payments
- Travel Plans
- Reviews
- Journals

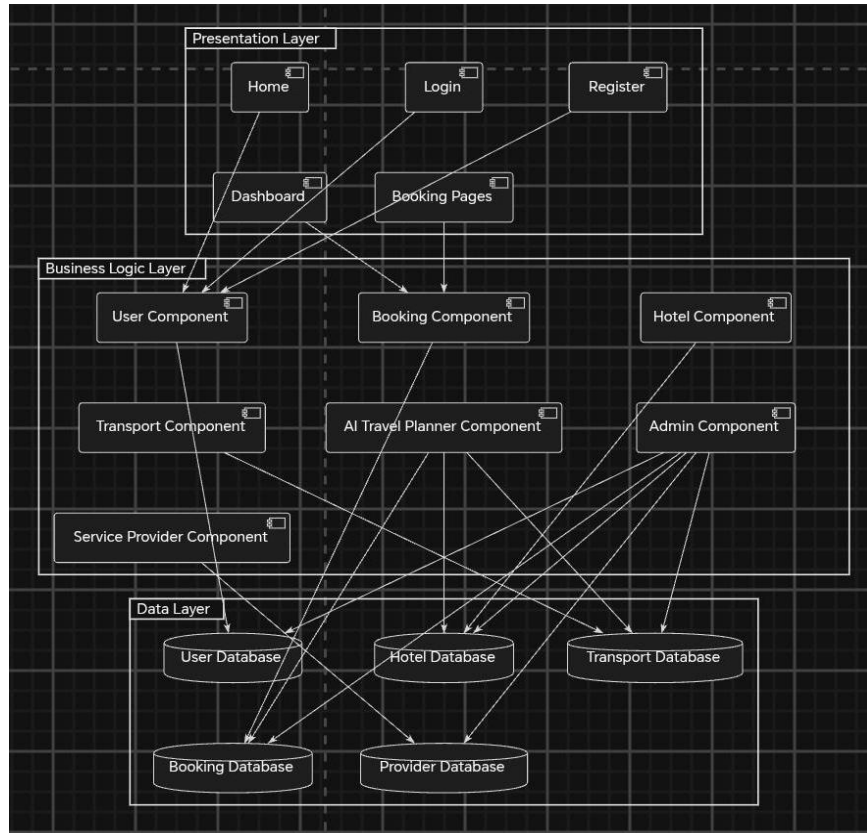


Figure 3.2: Top-Level Component Diagram

3.1.3 Component Elaboration

Each component of the system is described below.

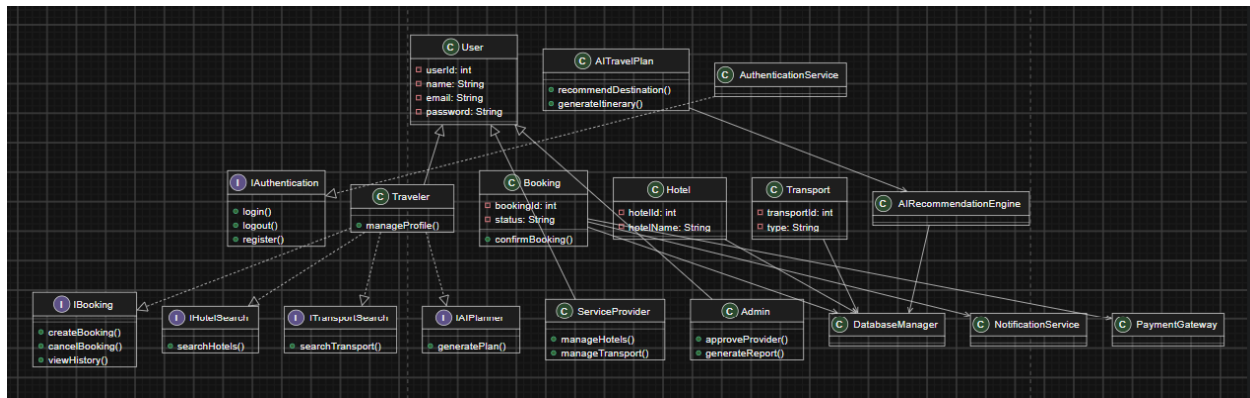
Component	Description
User Management	Handles authentication, profiles, travel history, and wishlists.
Booking Management	Processes transportation, accommodation, and restaurant reservations.

AI Module	Generates itineraries, recommendations, budget predictions, and travel partner suggestions.
Emergency Module	Provides SOS services, emergency notifications, and safety alerts.
Community Module	Supports travel blogs, reviews, discussion forums, and group travel.
Administration Module	Manages users, agents, reports, and system analytics.
External Services	Integrates maps, weather, payment, and AI services.
Database	Stores all application data securely.

Table 3.1: Component Elaboration

3.2 Component-Level Design

Component-level design provides a detailed description of the internal structure of each major component.



3.2.1 Elaboration of Design Components

User Management Component

Functions

- User Registration
- Login
- Password Encryption
- JWT Authentication
- Profile Update
- Wishlist Management

Inputs

- User Information
- Login Credentials

Outputs

- User Dashboard

- Authentication Token

Booking Component

Functions

- Search
- Book
- Cancel
- Reschedule

Inputs

- Destination
- Date
- Passenger Details

Outputs

- Booking Confirmation
- Ticket

AI Component

Functions

- Generate Itinerary
- Recommend Hotels
- Recommend Restaurants
- Estimate Budget

- Predict Fare

Inputs

- Budget
- Destination
- Duration
- Interests

Outputs

- Personalized Travel Plan

Emergency Component

Functions

- SOS
- Live GPS Sharing
- Emergency Notification

Outputs

- Alerts
- Emergency Response

Community Component

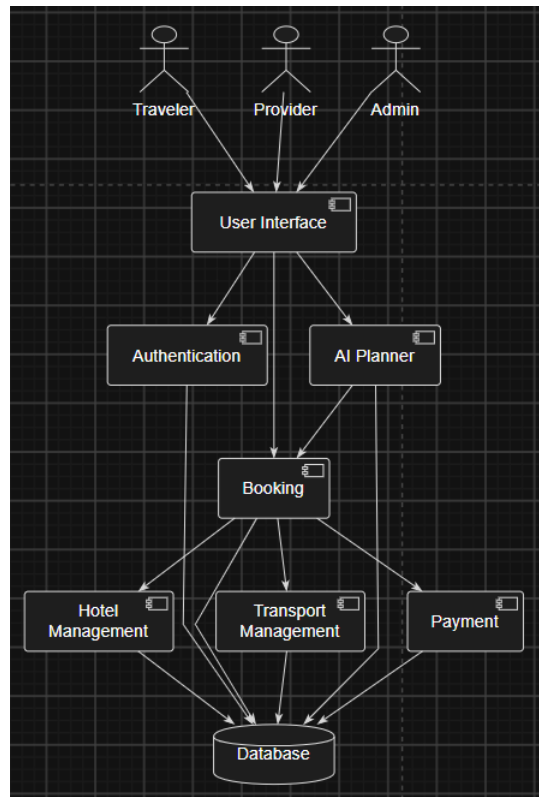
Functions

- Post
- Review

- Forum
- Journal

Outputs

- Shared Travel Content



3.2.2 Class Diagram

The proposed class diagram consists of the following classes:

- User
- TravelAgent
- Admin

- Booking
- Hotel
- Transportation
- Restaurant
- AIPlanner
- BudgetManager
- EmergencyService
- TravelJournal
- Review
- Payment
- Notification

These classes interact to perform booking, travel planning, emergency response, and user management.

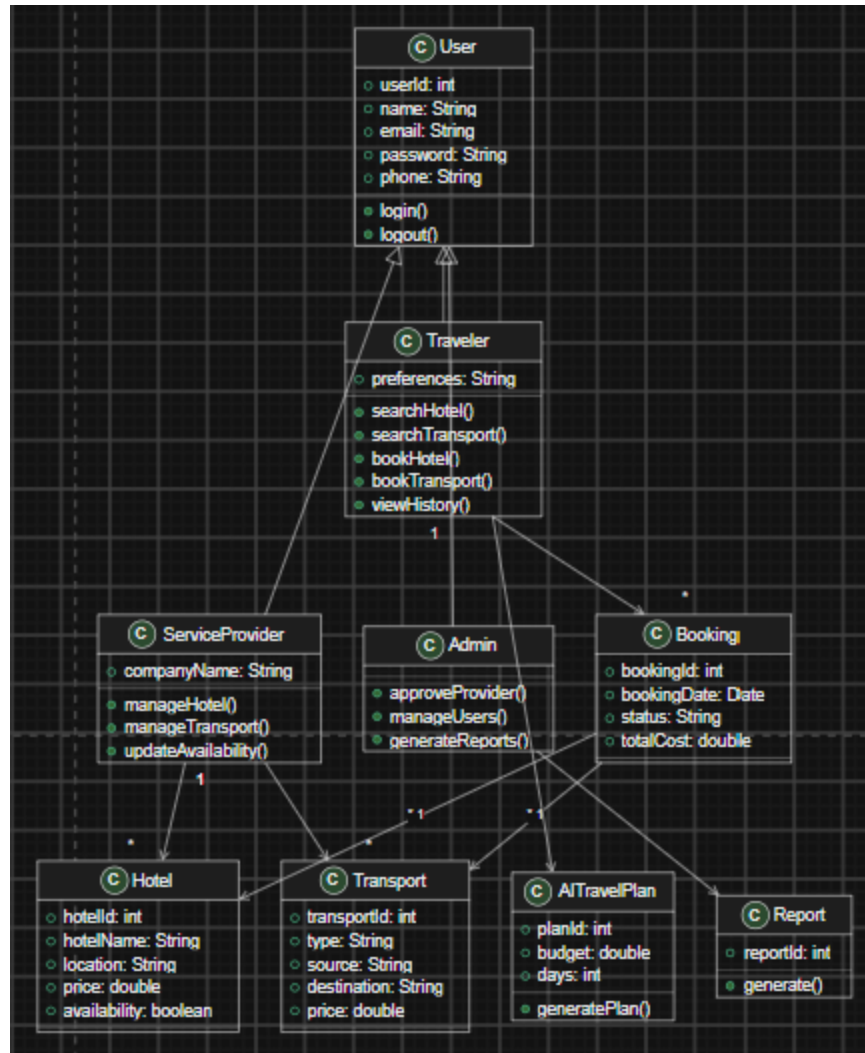


Figure 3.3: Class Diagram

3.2.3 Database Design

The database follows a relational structure to ensure consistency and efficient data management.

Main Tables

Table	Description
-------	-------------

Users	Stores user information
TravelAgents	Agent information
Admins	Administrator accounts
Bookings	Booking records
Hotels	Hotel details
Restaurants	Restaurant information
Transportation	Bus, Train, Flight, Ferry
TravelPlans	AI-generated itineraries
Reviews	User reviews
Payments	Transaction history
EmergencyContacts	Emergency information

Journals	Travel journals
----------	-----------------

Table 3.2: Database Tables

Entity Relationships

Main relationships include:

- One User → Many Bookings
- One User → Many Travel Plans
- One Booking → One Hotel
- One Booking → One Transportation
- One User → Many Reviews
- One User → Many Journals

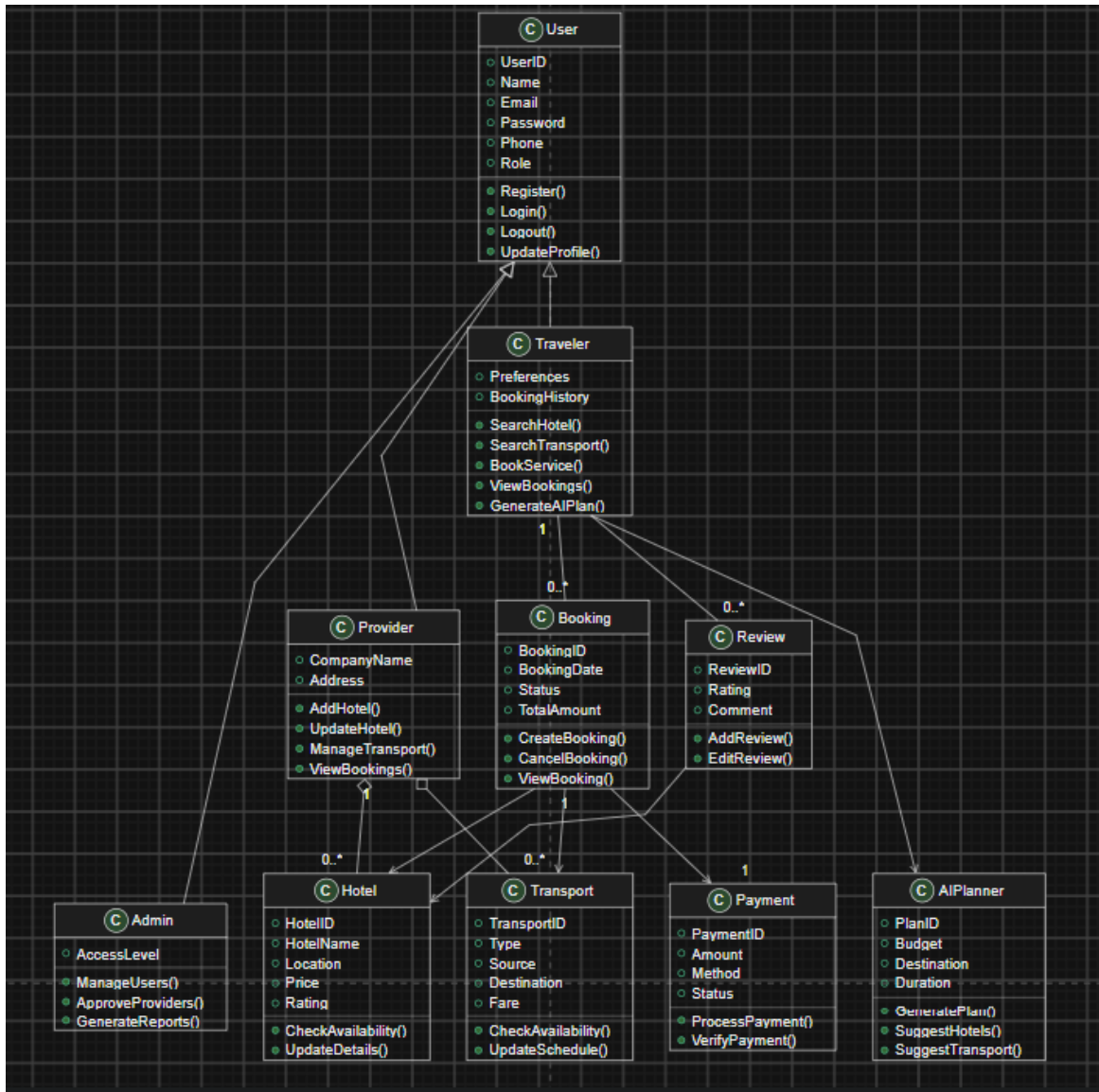


Figure 3.4: Entity Relationship Diagram (ERD)

3.3 User Interface Design

The user interface is designed to provide a clean, intuitive, and responsive experience for all users.

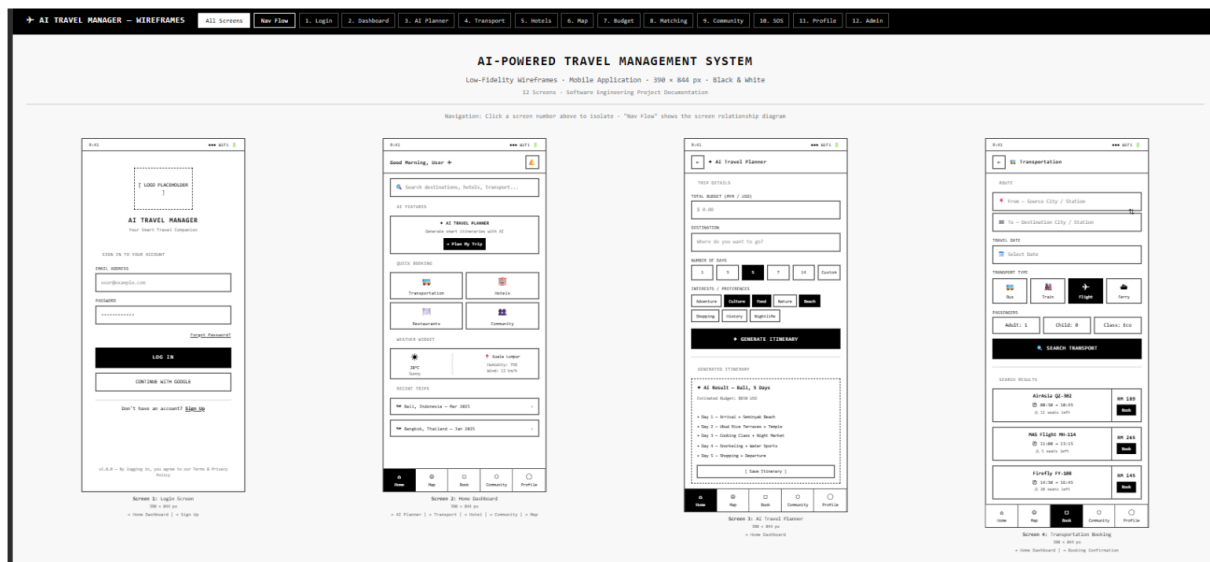
3.3.1 UI Wireframes

The system includes wireframes and prototypes for the following screens:

- Login & Registration

- Home Dashboard
- AI Travel Planner
- Transportation Booking
- Hotel Booking
- Restaurant Reservation
- Smart Budget Dashboard
- Interactive Map
- Emergency SOS
- Community Feed
- User Profile
- Admin Dashboard

These wireframes were designed using **Figma** before implementation.



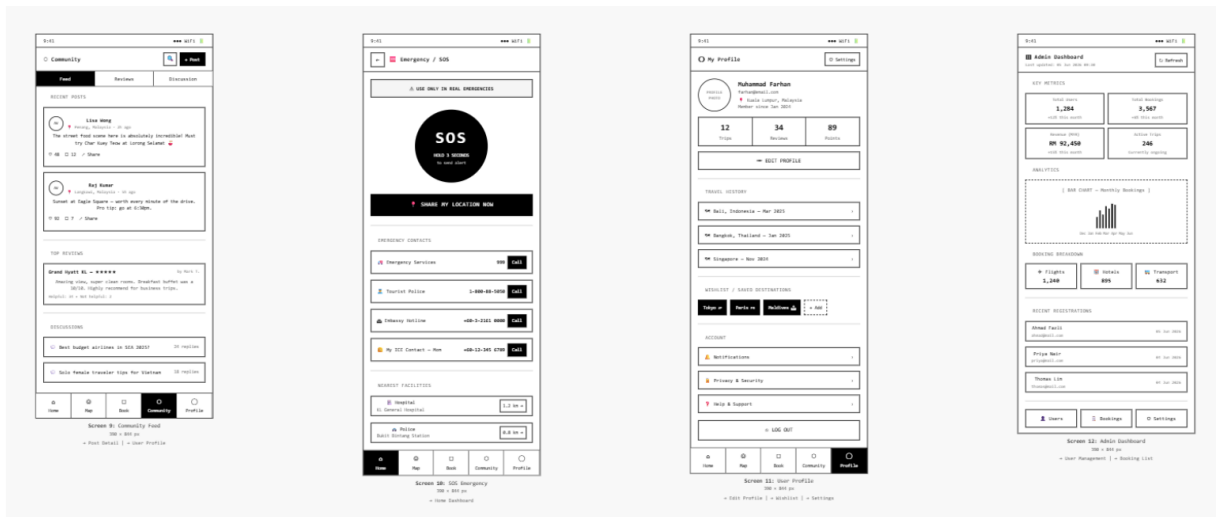
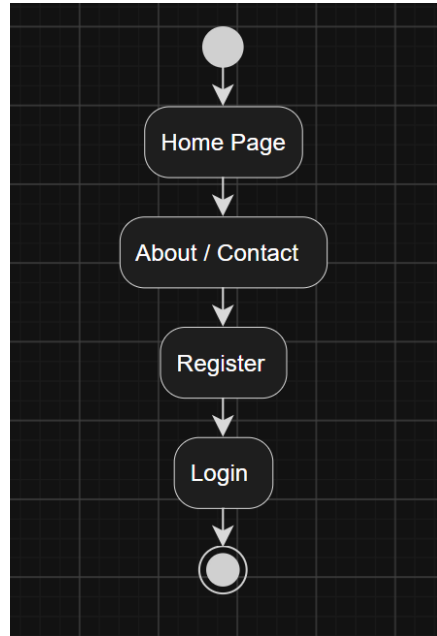


Figure 3.5: UI Wireframes

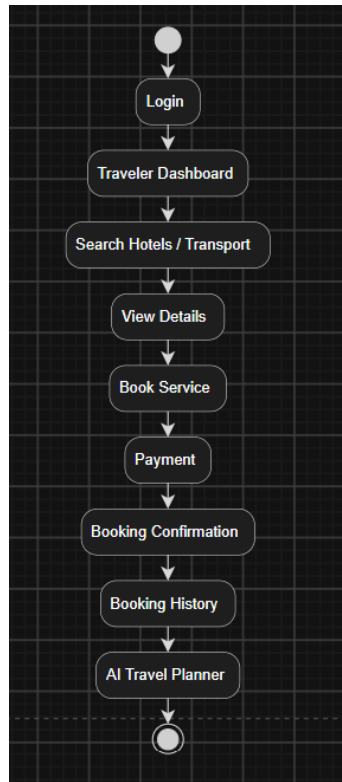
3.3.2 Navigation Flow

The navigation flow illustrates how users move through the application.



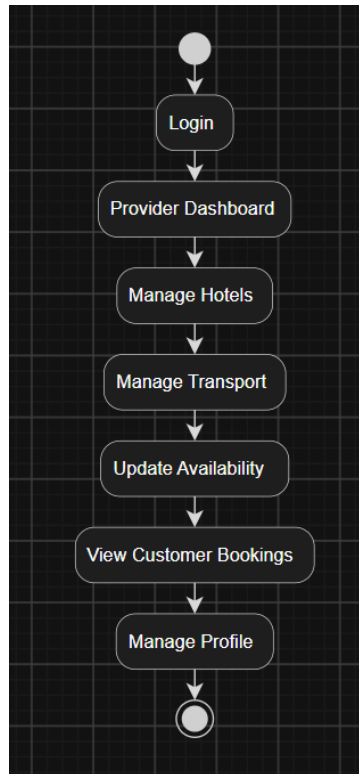
Guest User Navigation

A guest user first visits the Home page to explore the system. From there, the user can access the About and Contact pages to learn more about the platform and its services. To use the booking and AI planning features, the guest must create an account through the Registration page and then log in to access the system as a traveler.



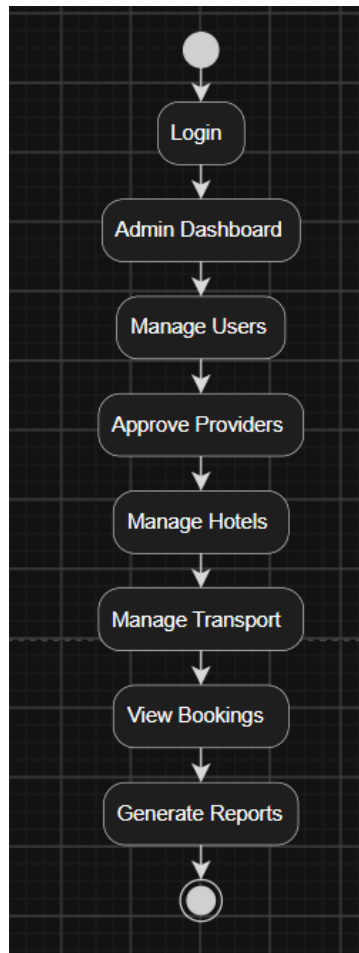
Traveler Navigation

After logging into the system, the traveler accesses the dashboard to search for hotels or transportation services. The traveler can view detailed information about available services, make bookings, and complete payment if required. After successful booking, the confirmation is displayed, and the booking details are stored in the booking history. The traveler can also use the AI Travel Planner to receive personalized travel recommendations.



Service Provider Navigation

The service provider logs into the system and accesses the provider dashboard. From the dashboard, the provider can manage hotel and transportation services, update availability, and monitor customer bookings. The provider can also update profile information whenever necessary.



Administrator Navigation

The administrator logs into the system through the admin portal and accesses the administrator dashboard. From there, the administrator manages users, approves service providers, oversees hotels and transportation services, monitors bookings, and generates reports to support system management and decision-making.

Chapter 4: Implementation

This chapter describes the implementation of the AI-Powered Travel Management System, including the project structure, technologies used, and development approach. The system follows a modular architecture to improve maintainability, scalability, and ease of development.

4.1 Code Structure Overview

The project is divided into two main parts: the **frontend** and the **backend**.

The frontend is responsible for the user interface, while the backend handles business logic, database operations, authentication, and communication with external APIs.

Project Structure



Figure 4.1: Project Folder Structure

4.2 Technology Stack

Layer	Technology
Frontend	React.js, HTML5, CSS3, JavaScript
Backend	Node.js, Express.js
Database	MongoDB
Authentication	JWT
Version Control	Git & GitHub
UI Design	Figma
Deployment	Vercel, Render
Maps	Google Maps API
Weather	OpenWeather API

AI	OpenAI API / ML Models
----	------------------------

Table 4.1: Technologies Used

4.3 Module Implementation

The system was implemented using independent modules to improve scalability and maintainability.

User Management Module

Implemented Features:

- User Registration
- Login
- Authentication
- Profile Management
- Travel History
- Wishlist

Transportation Module

Implemented Features:

- Search Transportation
- Seat Selection
- Booking
- Cancellation

- Rescheduling

Accommodation Module

Implemented Features:

- Hotel Search
- Room Booking
- Availability Checking
- Reviews

Restaurant Module

Implemented Features:

- Restaurant Search
- Reservation
- Food Pre-order

AI Travel Planner

Implemented Features:

- Personalized Itinerary
- Budget Estimation
- Fare Prediction
- Hotel Recommendation
- Restaurant Recommendation
- Travel Partner Matching

Emergency Module

Implemented Features:

- SOS Button
- Live Location Sharing
- Emergency Contact Notification

Community Module

Implemented Features:

- Travel Feed
- Discussion Forum
- Reviews
- Travel Journal

Admin Module

Implemented Features:

- User Management
- Agent Verification
- Complaint Handling
- Analytics Dashboard

4.4 GitHub Repository

The complete source code is maintained using Git and GitHub to support version control, collaboration, and project management.

Repository URL: <https://github.com/Sadia-23/travel-management-system>

Chapter 5: Deployment

This chapter describes the deployment process of the AI-Powered Travel Management System and the platforms used to make the application accessible online.

5.1 Deployment Platform

The system is deployed using cloud-based hosting platforms to ensure high availability and easy accessibility.

Component	Platform
Frontend	Vercel
Backend	Render
Database	Aiven-MySQL
Source Code	GitHub
AI API	Grok

Table 5.1: Deployment Platforms

5.2 Deployment Process

The deployment process consists of the following steps:

1. Develop the application locally.

2. Push the source code to GitHub.
3. Create and configure the database in Aiven.
4. Import the database into Aiven using HeidiSQL.
5. Configure the backend to connect to the Aiven database.
6. Deploy the backend to Render.
7. Configure the required environment variables in Render.
8. Deploy the frontend to Vercel.
9. Connect the frontend with the backend API.
10. Test all modules after deployment.

5.3 Website URL

After successful deployment, the application can be accessed using the following URLs:

Frontend: <https://travel-management-system-neon.vercel.app/>

Backend API: <https://travel-management-system-nr87.onrender.com>

Roles (Traveler/Provider):

Email: {layla, samiha, maliha, putul, provider1, provider2}@test.com

Password: test123

To open admin panel: Type <frontend link>/admin

Admin Email: admin@test.com

Admin password: admin123

Note: The passwords have been centralized for the roles, Travelers and providers, as the password is one-way encrypted and there is no method for decrypting it. Thus, for better testing and improvement, the passwords have been kept the same.

Chapter 6: Conclusion

6.1 Learnings

The development of the AI-Powered Travel Management System provided valuable experience in software engineering practices, including requirement analysis, system design, implementation, testing, deployment, and teamwork. Throughout the project, modern web technologies, cloud deployment, UI/UX design principles, and AI-based solutions were explored and applied. The project also enhanced problem-solving, collaboration, and project management skills.

6.2 Limitations

Although the proposed system provides a comprehensive travel management solution, several limitations remain:

- Some third-party APIs may have usage restrictions or associated costs.
- AI recommendations depend on the quality and availability of user data.
- Offline functionality is limited compared to online services.
- Advanced AR/VR features are planned but not implemented in the current version.
- The system currently targets web users, with native mobile applications reserved for future development.

6.3 Future Plan

Several enhancements are planned to improve the system in future versions.

Future improvements include:

- Mobile applications for Android and iOS.
- AR-based navigation.
- VR destination previews.
- AI voice assistant with multilingual support.
- Smart city integration.
- Blockchain-based secure travel identity.
- Enhanced AI recommendation models.
- Integration with additional airlines, hotels, and transportation providers.
- Carbon footprint optimization.
- Wearable device integration for health and emergency monitoring.
- Travel Partner Matching
- Automatic Travel Journal Generation
- Lost Traveler Detection & Alert System
- Carbon Footprint Analysis
- Offline Navigation Support
- AI Weather-Based Itinerary Adjustment

These improvements will make the system more intelligent, secure, and user-friendly while expanding its capabilities to support future travel technologies.

References

Include all references used in the report, following the citation style required by your university. For example:

1. Ian Sommerville, *Software Engineering*, 10th Edition.
2. Roger S. Pressman & Bruce R. Maxim, *Software Engineering: A Practitioner's Approach*.
3. React Documentation.
4. Node.js Documentation.
5. MongoDB Documentation.
6. Express.js Documentation.
7. Google Maps Platform Documentation.
8. OpenWeather API Documentation.
9. Figma Documentation.
10. Draw.io Documentation.

Appendices

Appendix A: Survey Questionnaire

- What are your most common travel difficulties?

- Which booking method do you prefer?
- How important are AI-based travel recommendations?
- What safety concerns do you face while traveling?
- Which travel services do you use most frequently?

Appendix B: Interview Questions

- How do you currently plan your trips?
- What problems do you face while traveling?
- Which features would you like in a travel management application?
- What safety measures are important to you?
- Would AI-generated travel plans improve your travel experience?

Appendix C: UML Diagrams⁸⁵

- Use Case Diagram
- Activity Diagram
- Sequence Diagram
- Class Diagram
- ER Diagram
- Component Diagram
- Context Diagram

Appendix D: UI Wireframes

- Login Screen

- Home Dashboard
- AI Planner
- Transportation Booking
- Hotel Booking
- Restaurant Reservation
- Community Feed
- Admin Dashboard

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